The Local Origins of Business Formation: Entry as a Two-Stage Process

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Abstract

The business entry literature typically observes firms only at the first hire. We provide a new perspective using linked administrative microdata tracking the universe of U.S. business applications and their transition into employer firms. We model entry as a two-stage process: pursuit of a business idea (proxied by a business application) and implementation (transition). Results show these margins are distinct and associate differently with local conditions. While both margins matter, high-startup locations are characterized by high application intensity, whereas low-startup locations exhibit low transition rates, suggesting geographic disparities in entry arise from different dynamics at each stage of the entrepreneurial process.

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1 Introduction

Business startups provide important economic opportunities for entrepreneurs and play a central role in job creation, innovation, and productivity growth.[1] Yet startup activity varies dramatically across the United States: some regions are vibrant entrepreneurship hubs, while others are effectively startup deserts. Most empirical research observes firms only after they hire their first employee, in large part because available administrative and survey data rarely capture pre-hire entrepreneurial activity.[2] As a result, the nascent phase of entrepreneurship——when business ideas are formed, developed, tested, and often abandoned before the first worker is hired——remains largely empirically invisible and poorly understood. We overcome this limitation by linking new administrative data that follow the universe of U.S. business applications from initial filing to employer startup, thereby shedding light on the previously unseen pre-hire phase of entrepreneurship.

With this new data infrastructure, we can study startup formation as a two-stage process. In the first stage, potential entrepreneurs decide whether to pursue a business idea, which we proxy using applications for an Employer Identification Number (EIN). In the second stage, applicants decide whether to launch an employer firm by hiring their first worker. This framing naturally leads to a decomposition of startup intensity——measured as new employer startups per capita——into two margins: business-idea intensity (applications per capita) and the transition rate from application to employer startup. We show that both margins vary substantially across space, are only weakly correlated, and contribute independently to spatial differences in startup activity. Some locations generate many ideas but face barriers in converting them into employer firms, while others generate fewer ideas but convert them at high rates——differences that conventional data, which begin at first hire, cannot reveal.

Our two-stage view of entry stands in sharp contrast to the standard single-stage model implicit in much of the theoretical and empirical entry literature, in which startup formation is generally treated as a single decision observed at first hire: applications and startups are equivalent. This implies that geographic variation in applications and startups would move one-for-one, and that local conditions would affect both margins identically. Our framework and data allow us to test and reject this implication. Instead, we show that idea generation and implementation are distinct margins of entrepreneurship that respond differently to local economic and demographic conditions. By separating idea supply from

[1] See, e.g., Haltiwanger, Jarmin and Miranda (2013), Kerr, Nanda and Rhodes-Kropf (2014), and Decker, Haltiwanger, Jarmin and Miranda (2014).

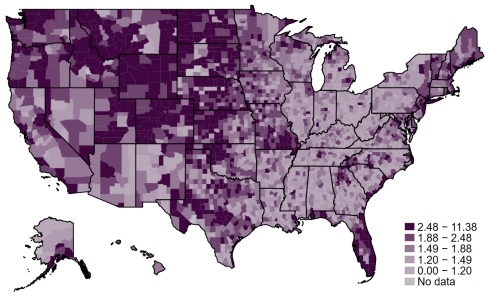
[2] The large literature includes Dunne, Roberts and Samuelson (1988), Davis, Haltiwanger and Schuh (1996), Haltiwanger et al. (2013), Alon, Berger, Dent and Pugsley (2018), Sedlacek (2020), Pugsley, Sedláček and Sterk (2021), and Karahan, Pugsley and Sahin (2024).

implementation frictions, our approach has important implications for how entry can be studied empirically and theoretically.

To guide the empirical analysis, we present a simple model of entry that formalizes the distinction between idea generation and implementation. In the model, potential entrepreneurs first decide whether to pursue an idea and then, after a gestational phase, decide whether to establish an employer firm. The model predicts that locations with high application rates need not have high transition rates, and that local conditions may affect these two margins in different directions. This framework motivates tests of whether observed spatial patterns in startup activity are driven by variation in idea supply, implementation frictions, or both.

Before turning to the confidential microdata, we present suggestive evidence from publicly available statistics to illustrate why observing both stages of entry matters. Using data from the Business Formation Statistics and Business Dynamics Statistics (BDS), Figures 1 and 2 show substantial cross-county variation in startups per capita, applications per capita, and transition rates.[3] Importantly, high-startup regions are not uniformly characterized by both high idea generation and high transition rates. For example, counties in Washington state exhibit high startup intensity primarily through elevated transition rates despite relatively low application rates, while in Florida high startup intensity reflects high application rates but comparatively low transitions. These contrasting patterns point to distinct dynamics in idea generation and implementation, motivating our use of confidential microdata to directly decompose these margins.

Figure 1: Startups per capita, by county

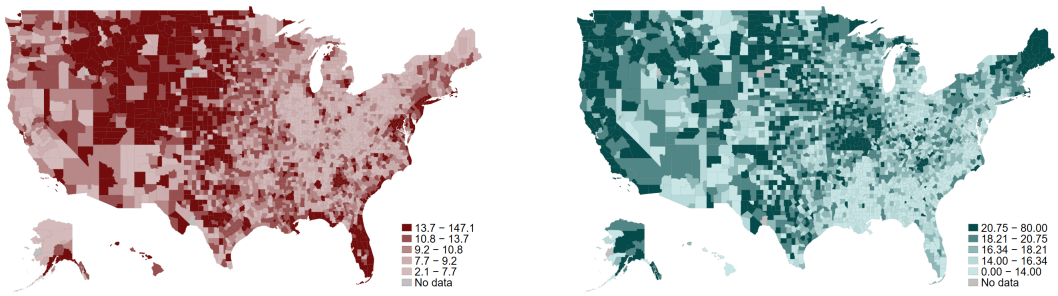


Notes: Average startups per capita (per 1,000 prime-age individuals) at the county level (2011-2016). See text for details.

Empirically, we use newly linked microdata from the Census Bureau's Business Formation

[3]Because publicly available county-level startup counts from the BDS are not directly linked to county-level application counts in the BFS, we realign timing for these figures by shifting startup counts back two years: startups observed in year $t+2$ are assigned to application year t (e.g., 2018 startups are mapped to 2016). This provides an approximate measure of application-to-startup transitions at the county level that is consistent with the timing we use in the BFS microdata.

Figure 2: Decomposing startups per capita, by county



(a) Applications per capita (b) Transition rate

Notes: Average applications per capita (left) and transition rates (right) at the county level (2011-2016). See text for details.

Statistics (BFS) and the Longitudinal Business Database (LBD) to follow the universe of EIN applications through their transition into employer firms.[4] Because all employer businesses are required to obtain an EIN and the LBD tracks all employer firms, these data allow us to observe every application that becomes an employer startup. To our knowledge, this is the first study to link the universe of business applications to subsequent employer formation nationwide in a way that enables spatially granular analysis of nascent entrepreneurship at the census tract level.[5]

Our analysis yields five main findings. First, both applications per capita and transition rates exhibit substantial geographic variation, and neither margin alone can account for spatial differences in startup intensity. Second, variance decompositions show that roughly two-thirds of tract-level variation in startups per capita reflects variation in applications per capita, while one-third reflects variation in transition rates, with little covariance between them. Third, high-startup locations tend to be characterized by high application intensity, whereas low-startup locations are more often constrained by weak transition rates. Fourth, local conditions relate differently to these two margins: household economic characteristics are more strongly associated with idea generation, while demographic characteristics are more strongly associated with transitions to employer status. Fifth, the relationship between the two margins of nascent entrepreneurship and demographic, economic, and business conditions frequently differ in both sign and magnitude. For example, higher median house values are associated with both higher applications per capita and higher transition rates.

[4]For details on the development of the BFS, see Bayard, Dinlersoz, Dunne, Haltiwanger, Miranda and Stevens (2018).

[5]Applications for EINs reflect nascent entrepreneurial activity, consistent with evidence from the Panel Study of Entrepreneurship Dynamics (PSED; see Reynolds, Carter, Gartner and Greene (2004)).

In contrast, a larger African American population share correlates with higher application per capita but lower transition rates, producing fewer startups per capita overall.

Because the relevance of idea generation and transition rates may differ across types of entrepreneurship, we separately examine high-tech and small-business-intensive (SBI) sectors, which differ markedly in entrepreneurial motivations, capital requirements, and outcome risks. Although these sectors differ in the relative importance of each margin for startup intensity and in how each margin relates to specific local conditions, the broad message remains the same: both margins exhibit substantial spatial variation, contribute meaningfully to startup intensity, and respond differently to local environments.

In SBI industries, patterns broadly mirror those in the pooled analysis. Similar to the pooled sample, applications per capita account for nearly three-quarters of the variation in SBI startup rates, while transition rates account for about one-quarter, with little covariance between the two. Household economic conditions are especially important in explaining variation in application rates, whereas demographic factors matter more for transition rates. Specifically, per capita income and house values are strongly positively associated with applications per capita, while African American population share is strongly negatively associated with transition rates; notably, this negative relationship with SBI startup intensity is even stronger than in the pooled analysis.

In high-tech sectors, four patterns stand out. First, activity is highly spatially concentrated: about half of tracts report zero applications over the sample period. Second, among tracts with activity, applications per capita explain 2.7 times more variation in startups per capita than transition rates, though both margins matter. Moreover, the two margins are negatively correlated with the covariance term accounting for a larger share of the variance in startup activity across tracts in comparison to the all industry and SBI decompositions. Third, demographic characteristics and household economic conditions dominate in accounting for variation across tracts. Applications per capita respond strongly and positively to the bachelor's-degree share and per capita income, while transition rates are highly negatively associated with African American population share and strongly positively associated with median house values.

Taken together, these results yield three contributions. First, by linking the universe of U.S. business applications to subsequent employer formation, we make the pre-hire phase of entrepreneurship empirically observable at a granular spatial level. Second, we develop a two-stage framework for entry that separates idea generation from implementation and generates testable predictions about how local conditions distinctly shape each margin. Third, using this framework, we provide the first tract-level evidence that spatial variation in startup activity reflects two distinct margins that are only weakly correlated, respond differently to

local economic and demographic conditions, and jointly determine startup intensity.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the theoretical framework. Section 4 describes the data. Section 5 analyzes the variance decomposition of startup rates. Section 6 examines the role of local conditions. Section 7 explores industry heterogeneity. Section 8 presents robustness checks and extensions. Section 9 concludes.

2 Review of literature

While the theoretical literature typically models idea creation and business entry as a single decision, empirical work has begun to explore phases of nascent entrepreneurship. Existing studies, however, tend to focus either on early-stage idea formation or on later firm outcomes, and rarely observe both phases at scale and across sectors and space. We advance this literature by using comprehensive administrative microdata to study idea generation and the transition of ideas into employer startups across locations and sectors. In doing so, we build on three strands of empirical work.

The first surveys nascent entrepreneurs. The Panel Study of Entrepreneurship Dynamics (PSED) is one such early effort (Reynolds et al., 2004; Reynolds, 2017), identifying about 5,000 individuals who have taken steps within the last year toward creating a venture but have not yet paid employees for more than three months.[6] More recently, Bennett and Chatterji (2023) and Bennett and Robinson (2023) field a voluntary, internet-based survey about nascent entrepreneurship, finding that about 30 percent of the 50,000 respondents had considered starting a business.[7] Both Reynolds et al. (2004) and Bennett and Robinson (2023) highlight that many prospective entrepreneurs invest significant time and resources in a "gestational" phase prior to launching their startup, and document that the majority of these individuals never open an employer business. Both studies also identify applying for an EIN as a key milestone. While these surveys offer rich individual-level insights, our use of rich administrative microdata enables three key improvements: full coverage of EIN applications, richer proxies for idea quality and viability, precise measurement of idea-to-startup transitions, and spatial and sectoral analysis of these dynamics.

A second strand of literature assesses the quantity and quality of entrepreneurship using state business registries (Andrews, Fazio, Liu, Guzman and Stern, 2018; Guzman and Stern, 2020). Guzman and Stern (2020) model high-impact outcomes (e.g., IPOs, acquisitions) as

[6]Wave I identified 4,000 and Wave II identified 1,500 individuals that satisfied this criteria.

[7]It is worth noting that the survey is re-weighted to be nationally representative in terms of key demographic characteristics.

a function of registration characteristics using data on state business registrations across 32 states between 1988 and 2014, while Fazio, Guzman, Liu and Stern (2021) examine the relationship between the surge in state business registrations during Covid-19 and zip-code level local conditions across eight states. While high-impact entrepreneurial outcomes are important, startups' role in local job creation and economic mobility makes it essential to understand all transitions to new employer businesses. Leveraging our detailed nationwide coverage and decomposition, we are able to examine the relationship between local conditions and application rates and transitions across all industries, as well as high-tech and small business sectors separately.[8]

Third, the spatial entrepreneurship literature investigates the role of local factors in shaping entrepreneurial activity. Glaeser, Rosenthal and Strange (2010a) review the urban economics perspective, noting both the role of entrepreneurship in urban growth and the influence of local characteristics on the supply of entrepreneurs. Studies such as Doms, Lewis and Robb (2010), Kerr and Kerr (2020), and Rosenthal and Strange (2003) explore factors including human capital, immigration, and industrial structure. We build on this work by separately identifying local drivers of idea generation and of transitions to employer status, and by documenting how these relationships differ across industries, suggesting avenues for research on sector-specific mechanisms such as entry costs, competitive dynamics, and workforce requirements.

3 From ideas to startups: a model of two-stage entry

To provide a conceptual foundation for our empirical decomposition of startup activity, we develop a simple model in which entrepreneurship unfolds in two phases corresponding to idea generation and the transition to an employer firm. Potential entrepreneurs differ ex ante in the quality of latent business ideas. They first decide whether to pursue an idea and then, after a gestational period during which additional information becomes available, decide whether to establish an employer firm. Because these decisions rely on different information and occur at different points in the entrepreneurial process, the model implies that idea generation and the transition to employer status may respond differently to local conditions.[9] The model thus provides a conceptual framework for our empirical analysis.

[8]The LBD and linked datasets permit similar analysis for high-impact outcomes, which we leave for future work.

[9]Pugsley et al. (2021) emphasize ex ante heterogeneity in a structural model that focuses on initial startup size and post-entry growth. We regard our approach as complementary: they emphasize heterogeneity at entry which they can measure at the time of first employment in the Longitudinal Business Database. Their model has a nascent entrepreneurship phase where some potential entrants incur fixed costs of entry, get a profitabilty signal, and choose not to enter. As a result, there is heterogeneity in initial size. Our observation

3.1 The model

Consider an economy where economic activity takes place in a large number of locations denoted by the set, $\mathcal{L}$. In each location $l \in \mathcal{L}$ there is a continuum of N_l individuals, each of whom has an idea, $\iota \in [0, \infty)$, for an employer business.[10] Higher values of ι indicate better (or higher quality) ideas in terms of expected return to an employer business. The distribution of ideas is given by the c.d.f. F_l (and density f_l).

An idea owner has to make an initial investment, $I_l > 0$, to pursue the idea. This investment involves both major planning tasks—such as estimating demand and costs, securing financing, understanding regulations, and seeking advice—as well as more mundane but necessary steps, including applying for an EIN and obtaining permits and licenses. Many of these tasks are related to addressing financial or labor-market related frictions, and can take time and resources to accomplish. Some ideas may thus take longer to transition to an employer business, or not transition at all, due to various local frictions involved.

Over the course of the investment, the idea owner sojourns in a state during which he observes a random signal, $V \in \mathbb{R}$, of the net value of an employer business. V can be thought of as an estimate of the idea owner's net payoff from the business—a scalar index of payoff-relevant factors for his business in location l, such as demand, entry costs (including any startup funding), various fixed and variable costs, the degree of competition, and his own productivity or ability.[11]

The value of V depends on the quality of the idea, summarized by the conditional distribution $G_l(V|\iota)$, which is known by potential entrants. A better idea (higher ι) may correspond to a higher V in a first order stochastic dominance sense. Each idea owner can choose to not pursue the idea and obtain a return of $R_l > 0$ (e.g. income from salary work or a nonemployer business). Importantly, R_l can reflect various labor market frictions. It can also reflect the disutility from being an employee rather than one's own boss (e.g., Hurst and Pugsley (2011)). For simplicity, we assume R_l and I_l do not depend on ι, though such

of applications often many quarters prior to the hiring of the first worker provides novel insights into this nascent phase permitting distinguishing between the application and transition stages of employer startup formation.

[10] We assume that ideas originate from the local population and are intended for businesses in the same location. While ideas can be aimed at other locations, we find empirically that planned business addresses substantially overlap with those of firms that ultimately form.

[11] The index V can thus include an initial signal of unknown cost or productivity parameter as in Jovanovic (1982) and the idea owner may continue to learn about that parameter after the business starts. Alternatively, V can depend on a known distribution of productivity as in Hopenhayn (1992), but one that depends on ι—reflecting pre-entry heterogeneity among potential entrants, in contrast to the identical potential entrants in Hopenhayn (1992). Chen, Croson, Elfenbeinand and Posen (2018) also explore learning dynamics in the pre-entry nascent entrepreneurship phase. A distinguishing feature of our model is that we specify that there are explicit costs associated with this pre-entry learning process. We think this is important in accounting for the spatial variation we observe in both idea creation and transitions.

dependence is plausible—for instance, individuals with a higher ι may earn higher wages. Higher investment can also improve the idea quality or change G_l, in the spirit of Ericson and Pakes (1995). While these considerations can be incorporated, we proceed with a simpler setup to illustrate our key points.

Each idea owner is "small" with respect to the local economy, and takes as given the local environment $\mathcal{E}_l = \{N_l, F_l(\cdot), G_l(\cdot|\cdot), R_l, I_l\}$. We do not study the determination of $\mathcal{E}_l$. In the background there is a set of spatial equilibrium conditions that ensure businesses and individuals optimize and have no incentive to move across locations, free entry holds in each location, and all markets clear. These equilibrium conditions pin down $\mathcal{E}_l$ for all $l \in \mathcal{L}$. Our focus is on the determination of what ideas are pursued and which ones turn into employer businesses in each location in a spatial equilibrium that generates $\mathcal{E}_l$.[12]

After observing V, an idea owner decides whether to start an employer business. The expected return from pursuing an idea is

$$\mathcal{V}_l(\iota) = E[\max\{V, R_l\}|\iota] = (1 - p_l(\iota))R_l + p_l(\iota)E[V|V \geq R_l; \iota] \tag{1}$$

where $p_l(\iota)$ is the probability that the pursued idea transitions to an employer business

$$p_l(\iota) = P(V \geq R_l|\iota) = 1 - G_l(R_l|\iota). \tag{2}$$

An idea owner will engage in nascent entrepreneurial activity (e.g., make an EIN application, among other steps) if $\mathcal{V}_l(\iota) \geq R_l + I_l$. Assume now that $\mathcal{V}_l(\iota)$ is increasing in ι—better ideas lead to higher expected potential return.[13] As long as $R_l + I_l < \mathcal{V}_l(\iota)$ for some ι, there exists a threshold (marginal idea) $\iota_l^* \in [0, \infty)$, such that all ideas in $[\iota_l^*, \infty]$ are pursued. The marginal idea satisfies

$$\mathcal{V}_l(\iota_l^*) = R_l + I_l, \tag{3}$$

and the mass of pursued ideas (business applications) per capita is

$$A_l = \frac{N_l \int_{\iota_l^*}^{\infty} f_l(\iota)d\iota}{N_l} = \int_{\iota_l^*}^{\infty} f_l(\iota)d\iota = 1 - F_l(\iota_l^*), \tag{4}$$

If $R_l + I_l > \mathcal{V}_l(\iota)$ for all ι, no idea is pursued ($A_l = 0$).

[12] In particular, we do not explicitly study the selection of individuals into a location, which determines the distribution of ideas, F_l. For instance, entrepreneurs may sort into locations based on their ideas and ability as in Nocke (2006). This sorting, however, may not be perfect since amenities and mobility frictions also factor in the determination of F_l.

[13] This is the case, for instance, if higher ι implies higher V in a first order stochastic sense, i.e. $G_l(\cdot|\iota)$ is decreasing in ι.

Startups per capita originating from applications is then

$$S_l = \frac{N_l \int_{\iota_l^*}^{\infty} p_l(\iota) f_l(\iota) d\iota}{N_l} = \int_{\iota_l^*}^{\infty} p_l(\iota) f_l(\iota) d\iota. \tag{5}$$

There are no startups ($S_l = 0$) if no idea is pursued ($A_l = 0$).

When $A_l > 0$, the (average) transition rate for applications is

$$T_l = \frac{S_l}{A_l} = \int_{\iota_l^*}^{\infty} p_l(\iota) f_l^*(\iota) d\iota = E[p_l(\iota) | \iota \geq \iota_l^*], \tag{6}$$

where $f_l^*(\iota) = \frac{f_l(\iota)}{1 - F_l(\iota_l^*)} = \frac{f_l(\iota)}{A_l}$ is the density of ideas conditional on application. Note that T_l is undefined when $A_l = 0$. By construction, the following holds

$$S_l = \begin{cases} A_l T_l & \text{if } A_l > 0, \\ 0 & \text{if } A_l = 0. \end{cases} \tag{7}$$

Expressions (1)-(7) hold in any spatial equilibrium. Now, we make explicit the fact that in equilibrium, each element of the local environment $\mathcal{E}_l$ will in general be a function of C, the collection of all relevant local characteristics or conditions, C_l, and conditions C_k in other locations $k \in \mathcal{L}$, $k \neq l$. These conditions can pertain to demographics, demand and costs, agglomeration, amenities, industrial composition, labor markets, laws and regulations, etc.[14] The individual elements of $\mathcal{E}_l$ need not depend on all elements of C. For instance, the distribution of ideas, F_l, may only depend on certain demographic characteristics of the population (e.g., age and education) in location l, but the initial investment, I_l, may, in addition, depend on financial frictions. The key variables of interest, A_l, T_l, and S_l, are then functions of the entire set of characteristics, C, because they depend on $\mathcal{E}_l$ by the definitions (3)–(6).[15]

Now consider the change in A_l, S_l, and T_l as a local characteristic or condition $c_l \in C_l$ changes from one location to another. Assume that A_l, T_l, and S_l are differentiable functions.

[14] Some of these conditions can be determined in spatial equilibrium (e.g., wages and rents), and some others (e.g., natural amenities) can be exogenous.

[15] A_l is a function of F_l and ι_l^*, which depends, through (3), on I_l, R_l, and G_l – which itself may be a function of local population (or local market size), N_l, that can matter for expected post-entry profit (see, e.g., Nocke (2006) and Asplund and Nocke (2006)). Similarly, S_l is a function of F_l, $p_l(\iota)$, and ι_l^*, which depend on I_l, R_l, and G_l. Finally, T_l is a function of S_l and A_l, and hence, a function of all conditions that the latter two depend on.

Then, for values of c_l for which $A_l > 0$, we have

$$\frac{dA_l}{dc_l} = -f_l(\iota_l^*)\frac{d\iota_l^*}{dc_l} + \int_{\iota_l^*}^{\infty} \frac{df_l(\iota)}{dc_l} d\iota, \tag{8}$$

$$\frac{dS_l}{dc_l} = -p_l(\iota_l^*)f_l(\iota_l^*)\frac{d\iota_l^*}{dc_l} + \int_{\iota_l^*}^{\infty} \left[\frac{dp_l(\iota)}{dc_l} f_l(\iota) + p_l(\iota)\frac{df_l(\iota)}{dc_l}\right] d\iota, \tag{9}$$

$$\frac{dT_l}{dc_l} = -p_l(\iota_l^*)f_l^*(\iota_l^*)\frac{d\iota_l^*}{dc_l} + \int_{\iota_l^*}^{\infty} \left[\frac{dp_l(\iota)}{dc_l} f_l^*(\iota) + p_l(\iota)\frac{df_l^*(\iota)}{dc_l}\right] d\iota. \tag{10}$$

Note that, for values of c_l for which $A_l = 0$, $\frac{dT_l}{dc_l}$ is undefined.[16]

Based on (8), A_l changes as c_l changes because not only the marginal idea ι_l^* changes, but also the distribution of ideas shifts. Similarly, (9) and (10) indicate that S_l and T_l change because the marginal idea, the distribution of ideas, and their transition rates all change. Observe that the sign of $\frac{dA_l}{dc_l}$ is in general unrestricted, and depends on the signs of $\frac{d\iota_l^*}{dc_l}$, and $\frac{df_l(\iota)}{dc_l}$.[17] Similarly, $\frac{dT_l}{dc_l}$ can be positive or negative, and its sign can differ from that of $\frac{dA_l}{dc_l}$. Note that by (6)

$$\frac{dT_l}{dc_l} = \frac{1}{A_l}\left(\frac{dS_l}{dc_l} - T_l\frac{dA_l}{dc_l}\right). \tag{11}$$

Thus, $\frac{dT_l}{dc_l}$ is negatively related to $\frac{dA_l}{dc_l}$, holding S_l constant. But $\frac{dS_l}{dc_l}$ can be positive or negative, and hence, the sign of $\frac{dT_l}{dc_l}$ is not the same as the sign of $\frac{dA_l}{dc_l}$ in general. From (8) and (9), the signs and magnitudes of $\frac{dA_l}{dc_l}$ and $\frac{dS_l}{dc_l}$ depend on how the idea distribution and the marginal idea change as c_l changes. However, the sign and magnitude of $\frac{dS_l}{dc_l}$ depends, in addition, on how the transition probabilities, $p_l(\iota)$, change. Depending on the nature of this change, the sign of $\frac{dT_l}{dc_l}$ can be the same as, or different from, the sign of $\frac{dA_l}{dc_l}$. Thus, without imposing additional structure on how F_l, G_l, and $p_l(\iota)$ respond to c_l, there is no general unambiguous sign comparison or magnitude ranking among $\frac{dA_l}{dc_l}$, $\frac{dS_l}{dc_l}$, and $\frac{dT_l}{dc_l}$ beyond the algebraic identity (11).

As an example, consider how A_l and T_l may vary with a local characteristic, such as the average level of educational attainment. Suppose that higher education is associated with higher idea quality: the idea distribution F_l under higher education first-order stochastically

[16] In interpreting the derivatives in (8)–(10), it is important to note that our analysis is reduced-form: we consider how an observed local characteristic c_l varies across locations, how this variation shifts the local environment $\mathcal{E}_l = \{N_l, F_l, G_l, R_l, I_l\}$ in a given spatial equilibrium, and how these induced changes in $\mathcal{E}_l$ translate into differences in A_l, S_l, and T_l across locations.

[17] Using (3), $\frac{d\iota_l^*}{dc_l} = \left[\frac{dI_l}{dc_l} + \frac{dp_l}{dc_l}\Big|_{\iota=\iota_l^*}(R_l - E_l) + p_l(\iota_l^*)\left(\frac{dR_l}{dc_l} - \frac{dE_l}{dc_l}\right)\right]\left(\frac{d\mathcal{V}_l}{d\iota_l}\right)^{-1}$, which depends on the rates of change in I_l, R_l, and G_l – the latter through the changes in p_l and $E_l = E[V|V \geq R_l; \iota_l^*]$. While $\frac{d\mathcal{V}_l}{d\iota} > 0$ by assumption, and $R_l < E_l$ by the definition of E_l, the rest of the terms cannot be signed without further restrictions. Similarly, $\frac{df_l(\iota)}{dc_l}$ can be positive or negative.

dominates the distribution under lower education. All else equal, (4) then implies that A_l is higher. Furthermore, (5) implies that S_l is also higher, as long as $p_l(\iota)$ is increasing in ι, i.e., better ideas transition with higher likelihood. Higher education may also directly raise transition probabilities by increasing the signal V. In terms of the signal distribution, this may correspond to a downward shift in $G_l(\cdot \mid \iota)$—a stochastically higher V—which increases $p_l(\iota) = 1 - G_l(R_l \mid \iota)$ for every ι. In this case, (5) implies a higher S_l, but the effect on A_l is only indirect through changes in the marginal idea. At the same time, the marginal idea ι_l^* also depends on G_l. For example, if greater competition accompanies higher education levels, expected profitability may fall, corresponding to an upward shift in $G_l(\cdot \mid \iota)$—a stochastically lower V. This raises the cutoff ι_l^* and lowers A_l. Depending on the signs and magnitudes of shifts in the idea distribution F_l and shifts in G_l that affect profitability and transitions, $\frac{dA_l}{dc_l}$ and $\frac{dT_l}{dc_l}$ may have different signs and magnitudes.

3.2 Implications

The primary hypotheses that emerge from this framework are as follows. First, the nascent exploration phase and the actual startup phase (marked by hiring the first worker) are distinct. Second, transition rates will be less than one and vary with entrepreneurial characteristics and local conditions. Third, the relationship between entrepreneurial characteristics and local conditions may differ across these distinct stages of the entry process.

These hypotheses contrast with a one stage entry process. To illustrate, consider a simplified case in which there is no additional information in the second stage (i.e. the observed signal is the expected value, $V = \mathcal{V}_l(\iota)$). In this case, only potential entrants with $\mathcal{V}_l(\iota) \geq R_l + I_l$ apply and transition.[18] Thus, the decision to apply coincides with the decision to transition, implying $A_l = S_l$ for all l and $T_l = 1$ for all l with $A_l > 0$. Consequently, there is no variation in T_l, and A_l and S_l exhibit identical variation across locations, including their relationships with local conditions.

Several special cases provide additional context for the analysis. Let ϵ_c^S, ϵ_c^A, and ϵ_c^T denote, respectively, the elasticities of startups per capita, applications per capita, and the transition rate with respect to an observable local condition (c). In a simple one-stage model where the decision to start and the decision to apply coincide ($S = A$), it follows that $\epsilon_c^S = \epsilon_c^A$ and $\epsilon_c^T = 0$. A related case arise when transitions are subject to independent random shocks uncorrelated with local conditions, so that $S < A$ and $T < 1$. Even in this case, $\epsilon_c^S = \epsilon_c^A$ and $\epsilon_c^T = 0$ continue to hold. Another case occurs when the hurdles for applying and transitioning are symmetric, in which case $\epsilon_c^A = \epsilon_c^T$.

[18] Note that the investment I_l could still be positive because of the costs associated with some deterministic tasks, such as obtaining a license or complying with regulations.

In taking this framework to the data, we evaluate two core implications of the model. First, we test whether observed spatial variation in startup activity is consistent with a one-stage or a two-stage entry process by examining whether idea generation and transition rates represent distinct margins that contribute independently to variation in startups per capita. Second, we examine whether local conditions relate differently to these margins, as predicted by the two-stage model. To do so, we estimate the elasticities ϵ_c^S, ϵ_c^A, and ϵ_c^T for a wide range of pre-determined local conditions. Our analysis is reduced-form and does not identify causal mechanisms; instead, it assesses whether empirical relationships across margins differ in magnitude, sign, and covariance, as implied by the two-stage entry process.

4 Data

4.1 Business applications

Our empirical analysis relies on micro data underlying the Census Bureau's Business Formation Statistics (BFS), which provide high-frequency measures of entrepreneurial activity derived from applications for EINs.[19] EINs are required for all U.S. employer businesses to file payroll taxes,[20] and the Census Bureau receives data on all new EIN applications from the IRS each week. To capture new business formation, the BFS excludes applications associated with trusts, estates, and other non-formation activities, using information reported on the application such as entity type and stated reason for applying.[21] One limitation is that sole-proprietor nonemployers are not required to obtain an EIN, though many choose to do so. As documented by Davis, Haltiwanger, Krizan, Jarmin, Miranda, Nucci and Sandusky (2009), sole proprietors with EINs report roughly three times higher revenues than those without, while non–EIN sole proprietors typically represent marginal activity. As a result, the BFS provides near-comprehensive coverage of economically significant business initiations in the United States.

The application form includes the name and address of the applicant and business, application week, business start date, reason for applying (e.g., hiring, banking), entity type, previous application for an EIN, principal industry, and planned date of initial wage payments, all of which serve as potential proxies for the underlying idea quality (ι). Business applications (BA) include all EIN applications in the BFS microdata, excluding applications associated with the purchases of an existing business or applications associated with changes

[19] For more information on the publicly available data, visit the BFS website.

[20] All new partnerships and corporations—whether employer or nonemployer—must also obtain an EIN. Nonemployers are businesses that file a business tax report with revenue but no employees.

[21] See Bayard et al. (2018) for details of the micro data, including the full set of filters.

in organizational type. Business applications with planned wages (WBA) are the subset in which applicants report a planned date for initial wage payments, indicating an intent to hire. Because our focus is on employer startups, most of our analysis centers on WBA.

Each business application includes a mailing address and, in some cases, a business address. Because EIN applications are filed at an early stage of the entrepreneurial process, this information often reflects the place of residence but can also be a distinct business address. Application addresses are geocoded to the Census county, tract and block levels.[22]

4.2 Business formations

To measure which applications transition into employer startups, the BFS links the confidential EIN application microdata to the Longitudinal Business Database (LBD).The LBD is the longitudinal version of the Census Bureau's Business Register and covers the universe of U.S. employer businesses. It records firm and establishment age, location, industry, number of employees, quarterly payroll, and EIN. Using the EIN as a common identifier, business applications are matched to the LBD to measure the incidence and timing of transitions to new employer businesses. The LBD identifies new employer startups as those firms with age equal to zero.

We focus on startups that occur within eight quarters of the application date. Figure 3 motivates this horizon: only one-fourth (BA) to one-third (WBA) of eventual transitions occur in the same quarter as the application, and by eight quarters-when transition rates begin to flatten-90 (BA) to 95 (WBA) percent of transitions have taken place.

Our main analysis focuses on spatial variation in WBA, the subset of applications in which nascent entrepreneurs indicate an intent to hire workers. These applications and their associated transitions align directly with our model, which features potential entrants aiming to start employer businesses. Our core sample covers all WBA applications from 2011–2016, encompassing more than 2.5 million applications and nearly 1 million resulting employer startups. Using these data, we measure each component of the identity (7), $S_l = A_l T_l$:

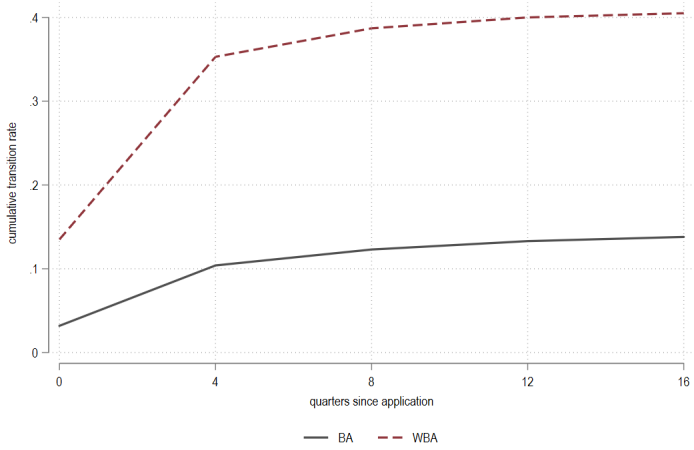
1. $\boldsymbol{S_l}$ **(startups per capita)**: startups that occur within eight quarters of the application date in location l divided by the prime age population (measured in thousands) of l.

2. $\boldsymbol{A_l}$ **(applications per capita)**: EIN applications in location l divided by prime age population (measured in thousands) of l.

3. $\boldsymbol{T_l}$ **(transition rate)**: EIN applications divided by startups in location l (S_l/A_l).

[22] Over 99 percent of applications are geocoded to the state and county level, while 85 percent are coded to the tract and block levels, see Bayard et al. (2018).

Figure 3: Cumulative transition rates for BA and WBA as a function of time since application



Notes: Cumulative transition rate of BA and WBA between 0 and 16 quarters after application. Due to disclosure considerations, the cumulative transition rates are calculated only for quarters 0, 4, 8 and 16. Transitions (startups) are defined as applications that transition to an employer business within eight quarters after application.

4.3 Assigning location and industry

We define location (l) using the geocoded county or tract on the EIN application. For applications that transition, we compare application address with the startup address. At the county level, over 90 percent of WBA transition in the same county as reported on the application, and nearly 80 percent do so at the tract level. In short, the application address and business address largely align at the county and tract levels.

We assign industry in a similar manner. Application industry codes are self-reported on the EIN form, while industry codes for transitions are based on additional sources, including from the Economic Census. For WBA startups, the application industry generally aligns with the startup's industry: 91 percent coincide at the 2-digit NAICS level, 88 percent at the 3-digit level, and 81 percent at the 4-digit level. In other words, industry information on the application is a reliable predictor of subsequent activity of the startup.

5 Variation in startups, ideas, and transition rates

5.1 County and tract-level statistics

To set the stage for our empirical analysis, Table 1 summarizes WBA startups per capita, applications per capita and transition rates at the county and tract levels. The average county and tract appear similar, with 0.71 and 0.91 startups per capita arising from 1.6 and

2.3 applications per capita that transition at a rate of 0.39 and 0.37, respectively.[23] What really distinguishes the tract-level data is the high level of dispersion. The coefficient of variation for startups per capita and applications per capita is roughly eight times larger at the tract level than at the county level, and about twice as large for transition rates. This pronounced dispersion reflects a small number of extremely active tracts, producing right-skewed distributions in which tract-level means substantially exceed medians: 0.91 versus 0.65 for startups per capita and 2.3 versus 1.7 for applications per capita.

Table 1: Wage startups, applications, and transition rates: summary statistics

	County			Tract		
	(1)	(2)	(3)	(4)	(5)	(6)
	Startups pc	Applications pc	Transition rate	Startups pc	Applications pc	Transition rate
Mean	0.708	1.572	0.389	0.905	2.264	0.368
SD	0.335	0.739	0.104	4.586	9.711	0.153
CV	0.473	0.470	0.267	5.070	4.289	0.416

Notes: Mean, standard deviation (SD), and coefficient of variation (CV) of WBA startups per 1,000 prime-age people (startups pc), WBA per 1,000 prime-age (20–64 years old) people (applications pc), and WBA transition rate (startups divided by applications) between 2011 and 2016, reported separately for county- and tract-level data.

5.2 Variance decomposition

While startups per capita can be decomposed into the product of applications per capita and the transition rate (equation 7), it is an open empirical question whether both margins contribute meaningfully to spatial variation in startup activity, or whether one margin dominates. To evaluate whether each margin is independently informative, we perform a variance decomposition that quantifies the separate contribution of applications and transition rates to spatial variation in startup intensity. For locations with $S_l > 0$, (7) implies:

$$\log S_l = \log A_l + \log T_l. \tag{12}$$

The variance of $\log S_l$ can be decomposed as[24]

$$Var(\log S_l) = Var(\log A_l) + Var(\log T_l) + 2Cov(\log A_l, \log T_l). \tag{13}$$

[23] As a robustness check, in the appendix we also include analysis of all applications (BA).

[24] With slight abuse of notation, we use model variables to refer also to their observed counterparts, treated as random variables.

The covariance term can, *a priori*, be either positive or negative, depending on how A_l and T_l vary across locations. Section 3.1 showed that A_l and T_l need not move in the same direction as local conditions vary.

This decomposition holds only when startups (and hence applications and transitions) are positive. At the county level this restriction is negligible, but at the tract level it is more relevant because some tract-years have zero startups. To address this, we aggregate over 2011–2016 when constructing S, A, and T, which minimizes zeros at the cost of time variation. With this aggregation, more than 99 percent of tracts have at least one startup (and thus at least one application) over our sample period.

Table 2 shows that both applications and transition rates are distinct contributors to spatial variation in startups per capita from WBA. At the county level, approximately 74 percent of the variance in startups per capita is accounted for by variation in applications per capita, 18 percent by transition rates, and 7.5 percent by the covariance. At the tract level, despite greater dispersion, the shares are similar: 66 percent from applications, 33 percent from transition rates, and 2 percent from covariance. The sizable contribution of transition rate variation suggests that startup activity cannot be explained by a simple one-stage model equating application and startup variation.

Table 2: Variance decomposition for wage startups per capita

	County			Tract		
	(1)	(2)	(3)	(4)	(5)	(6)
	Applications pc	Transition rate	2 × covariance	Applications pc	Transition rate	2 × covariance
WBA	0.744	0.181	0.075	0.659	0.325	0.016

Notes: Variance decomposition of log(WBA startups pc) into log(WBA applications pc) and log(WBA transition rate) for 2011–2016, reported separately for county- and tract-level data. "pc" denotes per capita (per 1,000 prime-age people).

Broadly, county- and tract-level descriptive statistics and variance decompositions tell a similar story. Therefore, going forward we focus on tract-level analysis, which better aligns with where applications originate and employer startups form, and provides a more granular link between local conditions and entrepreneurial activity.[25] This approach also allows us to exploit substantial within-county dispersion in startups per capita, applications per capita, and transition rates.

[25]Rosenthal and Strange (2003) and Jofre-Monseny, Marín-López and Viladecans-Marsal (2011) also exploit within-county and within metropolitan area variation, respectively, to measure the presence of agglomeration. They show that the mechanisms of agglomeration often operate at relatively short or within-county distances.

5.3 Ranking analysis

The variance decomposition quantifies the average contributions of applications and transition rates to spatial variation in startup activity but may mask important heterogeneity across the startup distribution. To examine whether the relative importance of these two margins differs between high- and low-startup tracts, Figure 4 presents a decile-based rank analysis.

We divide all tracts (nearly 72,000) into deciles based on their startup-per-capita rank. Within each decile, we compute average ranks for startups per capita, applications per capita, and transition rates. By construction, startup ranks increase linearly across deciles. Because tracts are ordered by startup intensity, both application and transition rate ranks also tend to rise, but the shapes of these relationships differ markedly across the distribution.

Figure 4 reveals a pronounced asymmetry between the upper and lower halves of the startup distribution. High-startup tracts are characterized by especially high application intensity, whereas low-startup tracts are characterized by especially low transition rates. This pattern is evident when comparing average ranks in the fifth decile to those in the top and bottom deciles. In the top decile, startup ranks are 2.1 times those in the fifth decile, while application ranks are even higher at 2.2 times, and transition rate ranks are only 1.3 times as large. By contrast, in the bottom decile, startup ranks are one-ninth of those in the fifth decile, application ranks are 1.6 times lower, and transition rate ranks are dramatically lower—5.4 times smaller.

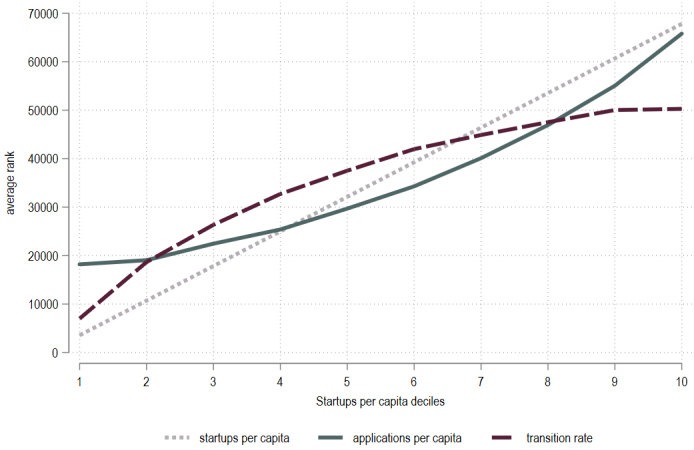
Taken together, these patterns indicate that strong startup regions distinguish themselves primarily through elevated application activity, while weak startup regions are distinguished by especially low transition rates. This suggests that startup deserts arise in part because applications in these areas are less likely to successfully transition into operating startups, highlighting the importance of local barriers to the transition process.

6 The role of local conditions

6.1 Empirical approach

Having established that applications per capita and transition rates each contribute meaningfully and distinctly to spatial variation in startups per capita, we now examine how these margins relate to observable local characteristics. Existing empirical work documents strong correlations between local conditions and startup activity. It is therefore natural to expect that local conditions may relate differently to applications and transition rates, helping to explain why each margin matters independently. To investigate these relationships, we

Figure 4: Tract-level rank analysis



Notes: Average ranks by decile for WBA startups per capita, applications per capita, and transition rates between 2011 and 2016 using at the tract-level data. "pc" denotes per capita (per 1,000 prime-age people).

estimate tract-level regressions using data from 2011–2016:

$$LHS_{lt} = \mathbf{C}'_{l,t-1}\boldsymbol{\beta} + f_{zt} + \epsilon_{lzt}, \tag{14}$$

where, in our baseline analysis, LHS_{lt} denotes one of three outcomes of interest—startups per capita (S_{lt}), applications per capita (A_{lt}), or transition rates (T_{lt}) in tract l in year t. $\mathbf{C}_{l,t-1}$ is a vector of lagged tract-level characteristics, f_{zt} are county-by-year fixed effects, and ϵ_{lzt} is the error term. The covariates are averages of the local conditions for the five-year interval ending one year prior to the year of the outcome variable.

Because the distributions of startups per capita and applications per capita contain zeros and are very right-skewed, we transform them using the Davis-Haltiwanger-Schuh (DHS) transformation, a second-order approximation of the log difference that accommodates zeros. Transformed startups per capita ($\widetilde{S}_{lzt}$) and applications per capita ($\widetilde{A}_{lzt}$) are expressed as deviations from their grand means, making them interpretable as the percent deviation of tract-year values from the overall average.[26] The transition rate (T_{lzt}) is untransformed and constrained to [0,1]. Both startups per capita and transition rate are conditional on $A_{lzt} > 0$. In our baseline analysis this is a minor restriction because 92.5 percent of tract-years have positive applications.[27]

[26]The DHS transformation is scale-free and avoids issues associated with the inverse hyperbolic sine transformation discussed by Chen and Roth (2023). See Appendix for elasticity interpretations.

[27]In the appendix we also report the results of regressions for two additional left hand side variables associated with applications: an indicator of positive applications (extensive margin) and applications per

Our selection of local conditions (**C**) is guided by insights from the theoretical and empirical literature on entrepreneurship, firm entry, and growth. Prior research emphasizes the importance of entrepreneur demographics such as age, education, and race (Doms et al., 2010; Fairlie and Miranda, 2016; Kerr and Kerr, 2017; Azoulay, Jones, Kim and Miranda, 2020, 2022; Bennett and Robinson, 2023), alongside their labor market prospects (Poschke, 2013; Choi, 2017; Dillon and Stanton, 2017; Hombert, Schoar, Sraer and Thesmar, 2020; Gottlieb, Townsend and Xu, 2022), and household wealth and access to finance (Evans and Jovanovic, 1989; Hurst and Lusardi, 2004; Adelino, Schoar and Severino, 2015; Hurst and Pugsley, 2017). In addition, research emphasizes the role of local economic conditions and characteristics of existing firms (Rosenthal and Strange, 2003; Glaeser, Kerr and Ponzetto, 2009, 2010b). We group local conditions of interest into three categories:

- **Demographic characteristics**: median age, education, race, ethnicity, and foreign-born population share.
- **Household economic conditions**: per capita income, unemployment rate, and median home values.
- **Incumbent firm characteristics**: share of employment in young firms, share of employment in large firms, average firm size, and 3-digit NAICS employment shares.

In addition to these three groups of variables, we include a measure of the tract's business activity—commercial share, defined as business employment divided by the sum of business employment and population. This variable ranges from zero to one, with higher values indicating more commercial activity in a location. We include it to capture differences across tracts arising from factors such as zoning (residential versus commercial), access to transportation and the availability of commercial real estate. In addition, this covariate captures potential local agglomeration effects emphasized by Rosenthal and Strange (2003).

We measure tract-year variables using data from the American Community Survey (ACS) and LBD (see Table A.1 for details). ACS variables are measured over five-year periods, and to be consistent with the ACS, the LBD covariates are constructed as five-year averages using the same time intervals. For both datasets, the five-year window corresponds to the period ending one year prior to the outcome year.

Including county-by-year fixed effects (f_{zt}) controls for broader market, regulatory, and labor market conditions that vary at the county level or higher. As a result, the covariates **C** capture within county-by-year variation across tracts. Table 3 provides summary statistics for the regression variables.

capita conditional on positive applications (intensive margin).

Our analysis does not identify causal mechanisms. While the use of lagged covariates mitigates concerns about simultaneity, unobserved factors may still influence both local conditions and entrepreneurial outcomes. Accordingly, our empirical objective is to continue our assessment of the model's implication that the margins of entry are distinct. We do so by estimating the elasticities of applications and transition rates with respect to local conditions and testing whether these elasticities differ across margins. This allows us to assess whether local environments relate differently to each stage of the startup formation process.

6.2 Baseline regression results

Before turning to our evaluation of specific local demographic, household, and incumbent firm characteristics, it is useful to benchmark how much variation can be explained by broader market conditions. We examine the explanatory power of county-by-year fixed effects alone, which absorb the market, regulatory, and labor environments shared by all tracts within a county-by-year. As shown in Table 4, these broader county-level factors account for less than one-quarter of tract-year variation in each outcome—more variation is explained for applications per capita (about 21 percent) and much less for transition rates and startups per capita (about 11 percent each). Thus, while county-level conditions matter, substantial heterogeneity remains across tracts within the same county, underscoring the potential importance of tract-level conditions. Importantly, the differential contribution of county-by-year effects across applications, startups, and transitions further supports treating these outcomes as distinct margins.

Table 5, which reports baseline estimates linking startups per capita, applications per capita, and transition rates to local conditions, yields two insights. First, local conditions account for 22 percent of variation in startups per capita. Second, they are more informative for applications per capita (27 percent) than transition rates (4 percent).

To unpack how local conditions relate to startup intensity, and potentially operate differently through applications per capita and transition rates, we require a metric that permits comparisons across variables and outcomes. Coefficients in Table 5 do not permit such comparisons on their own. We therefore convert coefficients into elasticities (at the means) and then into standardized magnitudes. We compute magnitudes by multiplying each elasticity by the coefficient of variation of the corresponding covariate, yielding the percentage change in the outcome associated with a one–standard-deviation change in the covariate from its mean—see Appendix Section F for details.

Table 3: Baseline regression variable summary statistics

	(1) Mean	(2) SD	(3) CV
DHS(WBA startups pc)	-0.520	0.977	-1.879
DHS(WBA pc)	-0.371	0.782	-2.109
WBA transition rate	0.373	0.289	0.775
log(median age)	3.629	0.200	0.055
bachelors or higher share	0.278	0.184	0.665
some college share	0.287	0.079	0.277
African American share	0.137	0.223	1.628
Asian share	0.045	0.087	1.929
Hispanic share	0.153	0.211	1.376
foreign born share	0.123	0.137	1.114
log(per capita income)	10.140	0.455	0.045
unemployment rate	0.094	0.059	0.625
log(median house value)	11.970	1.371	0.115
share of emp in young firms	0.175	0.124	0.705
share of emp in large firms	0.063	0.160	2.545
DHS(avg firm emp)	-0.287	0.664	-2.309
commercial share	0.179	0.170	0.946

Notes: Mean, standard deviation (SD), and coefficient of variation (CV) for dependent and control variables used in baseline regressions. "pc" refers to per capita (per 1,000 prime-age people). "DHS" denotes the transformation from Davis et al. (1996).

Table 4: Baseline contribution of county-by-year fixed effects

	(1) DHS(Startups pc)	(2) DHS(Applications pc)	(3) Transition rate
R-squared	0.111	0.208	0.108
Obs	398,0000	430,000	398,000

Notes: Share of the variance in DHS(WBA startups per capita), DHS(WBA applications per capita), and WBA transition rates explained by county-by-year fixed effects in the baseline tract-year sample (2011–2016).

Table 5: Baseline regression results

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Transition rate
log(median age)	0.138***	0.126***	0.0301***
	(0.0261)	(0.0247)	(0.0056)
bachelors or higher share	0.179***	0.0447	0.0506***
	(0.0408)	(0.037)	(0.0104)
some college share	–0.308***	–0.161***	–0.0513***
	(0.0522)	(0.05)	(0.0116)
African American share	–0.267***	0.522***	–0.197***
	(0.0378)	(0.0402)	(0.00654)
Asian share	0.0729	0.079	0.0182
	(0.109)	(0.0924)	(0.0228)
Hispanic share	–0.181**	–0.0881	–0.018
	(0.0784)	(0.107)	(0.0138)
foreign born share	0.271**	0.387***	–0.0536***
	(0.109)	(0.142)	(0.0141)
log(per capita income)	0.152***	0.231***	–0.0152***
	(0.0184)	(0.02)	(0.00416)
unemployment rate	0.447***	0.563***	–0.0214
	(0.0589)	(0.0513)	(0.0145)
log(median house value)	0.213***	0.142***	0.0385***
	(0.0156)	(0.0154)	(0.00231)
share of emp in young firms	0.210***	0.176***	0.00455
	(0.027)	(0.0197)	(0.00648)
share of emp in large firms	–0.534***	–0.557***	–0.0217***
	(0.0238)	(0.0183)	(0.0047)
DHS(avg firm emp)	–0.183***	–0.148***	–0.0108***
	(0.00588)	(0.00555)	(0.00142)
commercial share	2.694***	2.458***	0.144***
	(0.0343)	(0.0336)	(0.00538)
Obs	398,000	430,000	398,000
R-squared	0.3079	0.4203	0.1453
Within R-squared	0.2218	0.2677	0.04173

Notes: Regressions use tract-year–level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people). The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Elasticities for the baseline regressions are reported in Table 6. Because startups per capita equal applications per capita multiplied by the transition rate, the estimated elasticities should approximately satisfy $\epsilon_c^S \simeq \epsilon_c^A + \epsilon_c^T$. This relationship is not exact due to the DHS transformation and the presence of zeros, but it holds reasonably well in practice: the approximation error is small and centered near zero across covariates.[28]

[28]The mean approximation error, defined as $\epsilon_c^S - (\epsilon_c^A + \epsilon_c^T)$, is -0.014 using all applications and -0.008

Table 6: Baseline regression elasticities

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Transition rate
log(median age)	0.138	0.126	0.081
bachelors or higher share	0.051	0.012	0.039
some college share	-0.088	-0.046	-0.040
African American share	-0.037	0.072	-0.072
Asian share	0.003	0.004	0.002
Hispanic share	-0.028	-0.014	-0.007
foreign born share	0.034	0.047	-0.018
log(per capita income)	0.152	0.231	-0.041
unemployment rate	0.041	0.053	-0.005
log(median house value)	0.213	0.142	0.103
share of emp in young firms	0.037	0.031	0.002
share of emp in large firms	-0.034	-0.035	-0.004
DHS(avg firm emp)	-0.183	-0.148	-0.029
commercial share	0.501	0.441	0.072

Notes: Estimated percent change in the LHS variable induced by a one percent change in the RHS variable. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

The elasticity framework provides further tests of the model's central implication that applications and transition rates represent distinct margins of entry. Pairwise tests of equality across elasticities are reported in Appendix Table C.4.[29] For most covariates, elasticities differ significantly across margins: applications and transitions differ for 11 of the 13 covariates, applications and startups for 7, and startups and transitions for 10. In all cases where elasticities differ in sign across margins, the differences are statistically significant. Together, these patterns show that local conditions relate differently to applications and transition rates, consistent with the model's distinct-margins prediction.[30]

While the elasticities and associated tests highlight the two-stage nature of entry, Table 7 quantifies the economic importance of these differences by converting elasticities into stan-

when ϵ_c^A is estimated on the sample with positive applications.

[29]Tests are conducted on the sample of tract-years with positive applications, which provides common support for startups and transition rates.

[30]Because elasticities are estimated, p-values reflect tests of equality across approximate elasticities.

Table 7: Baseline regression magnitudes

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Transition rate
log(median age)	2.663	2.470	1.563
bachelors or higher share	3.325	0.798	2.543
some college share	-2.411	-1.274	-1.069
African American share	-5.994	11.722	-11.664
Asian share	0.566	0.772	0.378
Hispanic share	-3.800	-1.789	-0.950
foreign born share	3.703	5.347	-1.960
log(per capita income)	7.843	11.966	-2.116
unemployment rate	2.591	3.313	-0.308
log(median house value)	16.891	11.445	8.168
share of emp in young firms	2.512	2.185	0.136
share of emp in large firms	-8.452	-8.908	-0.994
DHS(avg firm emp)	-24.614	-21.712	-3.901
commercial share	46.192	41.624	6.638

Notes: Estimated percent change in the LHS variable induced by the percent change in the RHS variable equivalent to a one standard deviation multiple of the mean. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

dardized magnitudes. The standardization constructs the percent change in the outcome variable induced by a one–standard-deviation change in a covariate expressed as a percent of its mean. These magnitudes provide direct guidance on not only whether the application and transition margins are distinct, but also on the relative quantitative contribution of local conditions to each margin.[31]

Table 7 reveals three broad patterns. First, separating applications per capita and transition rates clarifies the relationship between startups per capita and local conditions. Consistent with the variance decomposition, in which applications per capita account for a larger share of startup variation, the signs of the startup-per-capita relationships typically mirror

[31]The magnitudes differ from elasticities only by a common scaling factor for a given covariate for the sample with positive applications. This implies the p-value tests reported above also apply to statistical differences in magnitudes. The reported magnitudes in the main text use the sample with zero applications so there are modest differences relative to magnitudes for the positive application sample only

those of applications per capita. When both margins reinforce each other (e.g., median house value, average firm size and college educated share) the magnitude is larger on startups per capita than applications per capita alone. In other cases where the two margins offset each other (as in African American share and per capita income), the net effect depends on which margin dominates.

Second, even when applications per capita and transition rates reinforce each other (as they do for nine of the thirteen covariates) one margin often plays a substantially stronger role than the other. In most cases, effects are larger at the application per capita margin. For example, median house value is positively related to both outcomes, but the effect on applications per capita (11.4) is about 40 percent larger than on transition rates (8.3). Similarly, the negative association with average incumbent firm size is nearly five times stronger for applications per capita (–21.7) than for transition rates (–4.3). In contrast, the college-educated share (bachelor's or higher) has the opposite pattern, exerting more than triple the positive effect on transition rates compared to applications.

Third, when the two margins offset each other—as in four of the thirteen covariates—the pattern is strikingly consistent: applications per capita are positively related to the covariate, while transition rates are negatively related. In most cases, at least one coefficient, and often both, are statistically significant. In three of the four cases (per capita income, foreign-born share, and unemployment rate), the net effect on startups per capita is driven by the applications margin. In the remaining case, the net effect is driven by the transition margin: higher African American population share is associated with more applications, lower transition rates, and fewer startups.

To complement this variable-by-variable evidence, we next assess how much of the overall variation in startups per capita, applications per capita, and transition rates can be attributed to our three categories of local covariates—demographic, household and incumbent firm. We do so using the variance decomposition methodology in Hottman, Redding and Weinstein (2016) and Eslava, Haltiwanger and Urdaneta (2024). This decomposition attributes to each covariate both its direct contribution to the variance of the outcome and its indirect contribution through covariances with other covariates.[32] By construction this method yields a decomposition where all terms, including the residual, sum to one.

Table 8 reports the variance decomposition for the baseline regressions in Table 5. Among our three categories of local conditions, household and demographic characteristics contribute similarly to accounting for variation in startups per capita. The importance of these two categories can be rationalized by looking at applications per capita and transition rates

[32]Formally, the contribution of a covariate is given by the product of the estimated coefficient, its covariance with the dependent variable and the ratio of its standard deviation to that of the dependent variable.

Table 8: Baseline regression decomposition

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Transition rate
Groups			
Demographic	0.025	0.008	0.027
HH economic conditions	0.035	0.042	0.006
Incumbent firm characteristics	-0.011	-0.019	0.002
Commercial share	0.173	0.237	0.007
Categories			
Local conditions	0.222	0.268	0.042
Common market conditions	0.086	0.153	0.104
Residual	0.692	0.580	0.855

Notes: Decomposition of R^2 across variable groups for DHS(WBA startups pc), DHS(WBA pc), and WBA transition rates. See notes to Table 5.

separately. Household economic conditions, driven primarily by median house value and per capita income, emerge as the most important group of local covariates for explaining applications per capita.[33] Meanwhile, demographics, driven by African American share and college educated share, are more informative for transition rates.

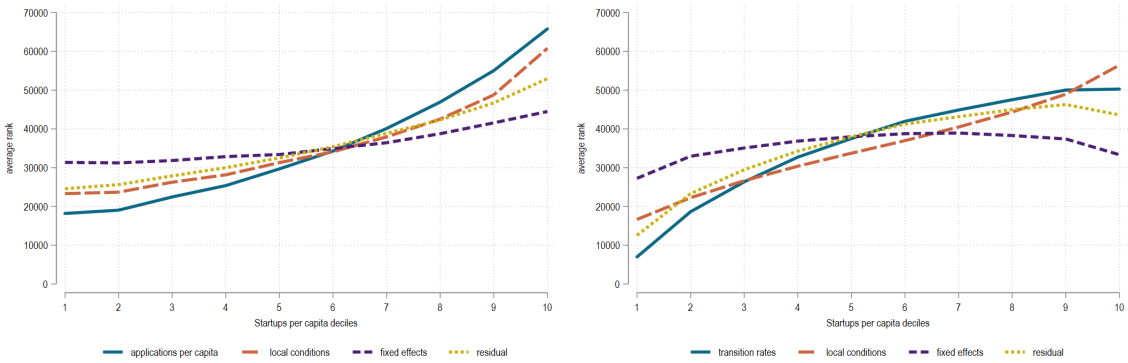
6.3 Ranking analysis revisited

To better understand how local conditions shape startup activity across the distribution of startup intensity, we revisit our ranking analysis to assess the extent to which differences in tract-level startup rankings reflect variation in applications per capita and transition rates. Formally, using our baseline regression models for applications per capita and transition rates, we construct predicted values for each outcome by isolating the contributions of (i) local covariates, (ii) county-by-year fixed effects, and (iii) the residual. We then average each predicted component over time at the tract level. Tracts are ranked according to each component, and the resulting ranks are averaged within each startup-per-capita decile.

Looking first at applications per capita (Figure 5a), rankings based on local covariates

[33]Surprisingly, the contribution of incumbent firm characteristics is negative, reflecting the role of covariance terms. The covariance contribution is evenly allocated across variables. It equals the product of the estimated regression coefficients and the covariance of the covariates. When two covariates are positively correlated but have opposite regression coefficient signs, their covariance contribution can be negative. For a covariate's total contribution to be negative, its direct contribution (squared coefficient times the variance) must be small relative to a negative covariance contribution.

Figure 5: Tract-level baseline rank analysis based on estimates



(a) Applications per capita

(b) Transition rate

Notes: Rank analysis based on WBA startups per capita at the tract level. "pc" refers to per capita (per 1,000 prime-age people). Plots depict average ranks by decile using observed applications per capita (left panel) and transition rates (right panel), along with corresponding predicted values based on local conditions, county-by-year fixed effects, and the residual.

closely track the raw series in Figure 4, particularly among high-startup tracts.[34] This alignment is somewhat surprising given the modest explanatory power of the regression model, but it highlights a key result: variation in applications per capita associated with observable local conditions closely tracks the ranking of startup intensity. In particular, local covariates account for the especially high application rates in the most startup-intensive locations. Quantitatively, predicted ranks based on local covariates mirror the raw data: the top-to-fifth decile average rank ratio is 2.2 in the raw data versus 1.9 based on observables. In contrast, county-by-year fixed effects and residuals are less predictive, yielding ratios of 1.3 and 1.5, respectively.

Turning to transition rates (Figure 5b), high (low) startup tracts also tend to exhibit higher (lower) predicted transition rates based on local conditions. County-by-year fixed effects show little connection to startup rankings, while the residual component plays a more prominent role, particularly in the lower tail of the distribution. Among the lowest-startup tracts, the fifth-to-first decile rank ratio is 5.4 in the raw data, compared with 2.1 for observables, 1.4 for county-by-year fixed effects, and 3.2 for the residual. These patterns suggest that while observables meaningfully contribute to rank differences in transition rates, unmeasured factors—such as idea quality and entrepreneur ability—captured in the residual are

[34] As a robustness check, we estimate the regression excluding the local commercial share to assess whether this covariate disproportionately drives the contribution of local conditions. The qualitative and quantitative patterns remain robust. For example, using local covariates, the ratio of the average rank of the 10th to the 1st decile is 2.6 in the baseline specification and 2.0 when the local commercial share is excluded.

especially important. Nevertheless, observable characteristics—such as the African American population share—help account for why low startup-intensive tracts exhibit especially low transition rates.

7 Industry heterogeneity

Because our baseline analysis pools all industries, it may obscure important differences across types of entrepreneurial activity. The relative importance of the idea generation and transition rate margins in explaining spatial variation in startup intensity, as well as how local conditions influence each, may vary across industries that differ in entrepreneurial motivations, capital intensity, and outcome risk.

To assess this heterogeneity, we examine nascent entrepreneurship separately in small business intensive (SBI) and high-tech sectors, which serve as illustrative cases. These industries differ in the type of ideas being pursued, the human and financial capital required at early stages, and the risks entrepreneurs must bear. SBI sectors typically involve locally oriented, lower-risk business opportunities with modest capital needs and more predictable demand patterns. Entrepreneurs in SBI industries are often motivated by non-pecuniary benefits of business ownership, and firms in these sectors typically exhibit low post-entry growth and little innovation. High-tech entrepreneurship, by contrast, is innovation-driven, skill-intensive, and highly scalable but also subject to substantial technological and market uncertainty. Consistent with this, these industries exhibit highly right skewed post-entry growth distributions (Decker et al., 2014), and have disproportionately contributed to innovation and productivity growth in recent decades. These sharp differences make the two sectors a useful pair for evaluating whether spatial variation in idea generation and transitions operates similarly across fundamentally different forms of entrepreneurial activity.[35]

Small business intensive (SBI) industries consist of forty 4-digit NAICS industries with the highest share of employer firms with fewer than 20 employees.[36] High-tech industries consist of the fourteen 4-digit NAICS codes identified by Hecker (2005) as STEM-intensive.[37]

[35] We conduct the same set of analyses for the SBI and high-tech samples as for the all-industry baseline sample. For brevity, we relegate the industry-specific ranking analyses to Appendix D.2, since the prevalence of zero-application and zero-startup tracts—especially in high-tech—makes rank-based comparisons less informative than in the baseline sample.

[36] This definition is related to, but distinct from, the Hurst and Pugsley (2011) (HP) classification, which ranks industries by the share of small firms it contributes to the overall economy. While there is considerable overlap, the HP ranking reflects industries with a large number of firms overall. The correlation between the HP ranking and the share of total firms an industry accounts for is 0.99, while the correlation between our SBI ranking and the HP measure is only 0.35, highlighting the distinction.

[37] This definition includes industries in manufacturing (NAICS 3254, 3341, 3342, 3344, 3345, 3364), information (NAICS 5112, 5161, 5179, 5181, 5182), and services (NAICS 5413, 5415, 5417).

For both SBI and high-tech industries, we construct measures of WBA startups per capita, applications per capita and transition rates following the approach outlined above.

Descriptive statistics: Table 9 reports summary statistics for SBI and high-tech industries. The table reports tract-level statistics pooled over 2011–2016 for startups per capita, transition rates, and three measures of application activity: applications per capita using all tracts; an indicator for whether a tract has any application activity in the industry (extensive margin); and the applications per capita conditional on having at least one application in the industry (intensive margin). The substantially higher level of startups per capita in SBI relative to high-tech is driven primarily by differences in applications per capita. These differences arise both at the extensive margin—96.4 percent of tracts have at least one SBI application over the sample period versus only 48.1 percent for high-tech application—and, at the intensive margin—0.984 SBI applications per capita compared with 0.515 for high-tech. In contrast, conditional on $A > 0$, transition rates are similar across the two sectors—0.395 for SBI and 0.371 for high-tech.

Table 9: Small business intensive industries and high-tech: summary statistics

	(1) Mean	(2) SD	(3) CV
Small Business Intensive			
Startups pc	0.396	1.122	2.831
Apps pc	0.697	2.190	3.144
Apps pc extensive	0.964	0.186	0.193
Apps pc intensive	0.984	2.240	2.276
Transition rate	0.395	0.244	0.619
High-Tech			
Startups pc	0.194	0.363	1.871
Apps pc	0.087	0.420	4.815
Apps pc extensive	0.481	0.500	1.039
Apps pc intensive	0.515	0.695	1.350
Transition rate	0.371	0.387	1.042

Notes: Mean, standard deviation (SD), and coefficient of variation (CV) of WBA startups per 1,000 prime-age people (startups pc), WBA per capita (applications pc), an indicator of positive WBA (apps pc extensive), WBA per 1,000 prime age people in tract-years with positive WBA (apps pc intensive), and WBA transition rate between 2011 and 2016, separately for applications filed as small business intensive and high-tech.

Table 10 reports pairwise correlations in startups per capita, applications per capita, and transition rates across the all-industry, SBI, and high-tech samples. The high correlation in startups per capita across all three samples indicates that tracts with high entrepreneurial activity in one sector tend to be active in others as well.[38] Unsurprisingly, business application and formation activity are especially highly correlated between the all-industry and SBI samples. In contrast, correlations involving high-tech are much stronger for applications per capita than for transition rates, suggesting more pronounced sectoral differences in the implementation stage of entry.

Table 10: Outcome variable correlations

	corr(all, SBI)	corr(all, high-tech)	corr(SBI, high-tech)
DHS(WBA startups pc)	0.870	0.479	0.601
DHS(WBA pc)	0.889	0.748	0.819
WBA transition rate	0.618	0.257	0.078

Notes: Pairwise correlations in startups per capita, applications per capita, and transition rates across industry samples. Column (1) reports correlations between all industry and small business intensive industry samples, column (2) between all industry and high-tech industry samples, and column (3) between small business intensive and high-tech samples.

Variance decomposition: In Table 11, we use industry-specific variance decompositions to assess how the contribution of applications per capita and transition rates to startup variation differs across sectors. Because these decompositions rely on log transformations, they are restricted to locations with positive applications and startups—an especially stringent restriction for high-tech industries at the tract level. We therefore report results at both the tract and county levels, where this limitation is less binding, and interpret the tract-level high-tech results with appropriate caution.

Our results indicate that both applications per capita and transition rates contribute meaningfully to spatial variation in startup activity across industries.[39] For SBI, especially at the county-level, the results mirror the all-industry decomposition in Table 2. In contrast, the high-tech sector shows a striking negative covariance between applications per capita and transition rates, in sharp contrast to the weakly positive covariance in the overall economy and the modest covariance estimates for SBI. This pattern indicates substantially weaker spatial alignment between the two margins in high-tech than in other sectors.

[38] Correlations use DHS measures, including zeros.

[39] As a cross-check, we re-estimate the all-industry variance decomposition on the subset of tracts used in each industry-specific analysis (Table D.1), using the same sample of tracts included in each industry-specific analysis. The results remain consistent with Table 2, suggesting that differences in the high-tech results are not solely driven by the smaller number of tracts with positive high-tech activity. Still, this check does not account for the fact that only a subset of these tracts also have positive high-tech startups.

Several mechanisms could generate this sharp negative covariance in high-tech. High-tech ideas tend to cluster in innovation hubs where idea generation is abundant, but competition for specialized labor, financing, and infrastructure may depress transition probabilities. These hubs may also produce a broader mix of speculative, high-variance ideas, lowering average transitions. By contrast, locations with fewer high-tech applications may generate only their most viable ideas and face less competition for scarce implementation resources, resulting in higher transition rates conditional on application. This negative covariance—distinctive to high-technology startup activity—would be impossible to detect absent our decomposition into separate application and transition margins.

Table 11: Variance decomposition for startups per capita, by industry

	County			Tract		
	(1)	(2)	(3)	(4)	(5)	(6)
	Applications pc	Transition rate	2 × covariance	Applications pc	Transition rate	2 × covariance
SBI	0.724	0.235	0.041	0.822	0.363	-0.185
High-Tech	0.950	0.420	-0.370	1.429	0.529	-0.958

Notes: Variance decomposition of log(startups pc) into log(applications pc) and log(transition rate) for 2011-2016, separately for county- and tract-level data. Row 1 reports results for the small business intensive and row 2 for the high-tech samples. Startups are defined as applications that transition to an employer business within eight quarters after application. "pc" refers to per capita (per 1,000 prime-age people).

Regression results: To assess whether the relationships between local conditions and the two margins of nascent entrepreneurship vary meaningfully across sectors, we next estimate separate models for SBI and high-tech industries. This approach allows us to evaluate whether startup activity across sectors is related similarly to local conditions, or whether SBI and high-tech exhibit distinct patterns. Because of the prevalence of zero-application tracts in high-tech, we supplement our three outcomes with two additional measures: an indicator for whether a tract-year has any applications in the industry (the extensive margin) and applications per capita conditional on having at least one application in the industry in that tract-year (the intensive margin). The regression-implied magnitudes are reported in Tables 13 and 14, with full regression results and elasticities in Appendix Tables D.4, D.5, D.7, and D.8.[40]

Five key insights emerge. First, as with the baseline results, startups per capita, applica-

[40]The estimated mean of the approximation error (given by $\epsilon_c^S - (\epsilon_c^A + \epsilon_c^T)$ across all all covariates is -0.024 when using all applications and -0.007 when for estimated ϵ_c^A we use the elasticities based on the sample with positive applications for SBI. The analogous statistics for HT are -0.008 and -0.011, respectively.

tions per capita, and transition rates exhibit distinct relationships with local conditions—see p-value test results in Tables D.6 and D.9.

Second, there is substantial overlap in the local conditions most strongly associated with startup activity across SBI and high-tech sectors, mirroring the baseline results. Table 12 summarizes this overlap by reporting the correlations between estimated elasticities across the baseline, SBI, and high-tech samples for each outcome. For a given outcome, we correlate the vector of elasticities obtained from regressions estimated separately on each sample using a common set of covariates. Correlations are uniformly high, with the exception of those involving high-tech applications per capita, which include both extensive and intensive margins. These weaker correlations reflect the large number of tracts without high-tech applications, which affects the estimated application elasticities. When we restrict attention to intensive-margin elasticities, correlations are high across all samples, including high-tech.

Table 12: Correlation of elasticities for baseline, SBI and high-tech samples

	DHS(startups pc)	DHS(apps pc)	Apps pc extensive	DHS(apps pc intensive)	Transition Rate
Corr(Baseline, SBI)	0.99	0.94	0.92	0.95	0.99
Corr(Baseline, HT)	0.92	0.37	0.65	0.84	0.70
Corr(SBI,HT)	0.93	0.63	0.81	0.96	0.75

Notes: For each outcome variable, the table reports the correlation of estimated elasticities across samples. Elasticities are obtained from separate regressions for each outcome (startups per capita, applications per capita, transition rates) and each sample (all industries, small business intensive, and high-tech), using a common set of covariates. Each cell reports the correlation between the vectors of elasticities for a given outcome across the indicated pair of samples.

This overlap is also evident in the specific covariates most strongly associated with startups per capita across sectors. Among the five covariates with the largest absolute elasticities for SBI (and for the pooled sample)—average firm employment, median house value, African American share, the share of employment in large firms, and per capita income—three (average firm size, African American share, and median house value) also appear among the top five for high-tech. The remaining two in the high-tech top five—college-educated share and Asian share—are far more relevant in high-tech than in SBI.

Importantly, these similarities are not confined to startups per capita but extend to the individual margins of entry. Across all fourteen covariates, only three exhibit a sign difference in at least one margin between SBI and high-tech: Asian share, which is negative for both margins in SBI but positive for both in high-tech; and some-college share and young-firm share, which show negative relationships with the transition rate in SBI but positive relationships in high-tech.

Table 13: Small business intensive regression magnitudes

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(pps pc intensive)	(5) Transition rate
log(median age)	2.850	-0.666	-3.352	6.821	1.178
bachelors or higher share	3.275	2.793	1.263	1.669	2.781
some college share	-3.158	-0.194	2.410	-4.377	-0.803
African American share	-10.536	5.047	1.465	5.025	-11.671
Asian share	-2.997	-1.929	0.000	-2.468	-1.058
Hispanic share	-4.828	-3.027	-0.138	-3.001	-1.957
foreign born share	2.660	6.684	4.679	0.716	-0.921
log(per capita income)	6.032	14.349	9.376	2.392	-1.508
unemployment rate	2.132	2.813	-0.250	4.750	-0.609
log(median house value)	13.099	13.621	7.979	3.753	7.813
share of emp in young firms	1.558	1.833	0.705	1.233	-0.065
share of emp in large firms	-6.063	-9.417	-3.309	-6.063	-0.466
DHS(avg firm emp)	-24.590	-18.924	1.467	-26.607	-4.287
commercial share	40.799	41.340	10.784	38.143	5.752

Notes: Estimated percent change in the LHS variable induced by the percent change in the RHS variable equivalent to a one standard deviation multiple of the mean. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Third, although the signs of relationships are fairly stable across industries, their magnitudes often differ sharply. One striking example is the college-educated share: in SBI, the magnitudes are modest (2.9 for applications per capita and 2.8 for transition rates), whereas in high-tech they are much larger, especially for applications per capita (18.3 and 3.8 for transition rates). Median house value shows a similar pattern. In SBI, the magnitudes resemble the baseline (12.9 for applications per capita and 7.8 for transition rates), but in high-tech they are far smaller (3.5 and 5.0, respectively).

Fourth, the covariates that offset each other in the baseline continue to do so across industry groups, though again with notable differences in magnitude. For example, African American share has much smaller effects in high-tech (1.1 for applications per capita and -6.9 for transition rates) than in SBI (4.8 and -11.7). Foreign-born share presents a particularly interesting case: although it is positively related to applications per capita and negatively to transition rates in both industries, the net relationship with startups per capita is positive in SBI (as in the baseline) but negative in high-tech.

Table 14: High-tech regression magnitudes

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
log(median age)	1.684	-6.821	-14.602	9.278	0.018
bachelors or higher share	6.257	16.758	33.849	1.185	3.839
some college share	-2.379	-1.579	-1.163	-7.074	0.970
African American share	-4.869	1.465	2.442	3.084	-6.817
Asian share	5.054	5.208	10.224	1.776	2.049
Hispanic share	-2.970	-6.192	-11.834	-1.544	-1.782
foreign born share	-1.666	6.016	12.143	-2.666	-2.749
log(per capita income)	2.008	13.727	25.848	-0.402	-1.857
unemployment rate	1.172	1.563	2.313	4.161	-0.703
log(median house value)	4.341	4.111	8.544	-1.447	5.030
share of emp in young firms	1.109	2.326	4.653	0.308	0.062
share of emp in large firms	-2.304	-3.054	-5.090	-3.648	0.000
DHS(avg firm emp)	-19.183	-4.694	-2.934	-28.180	-3.969
commercial share	24.194	18.920	31.029	30.910	2.669

Notes: Estimated percent change in the LHS variable induced by the percent change in the RHS variable equivalent to a one standard deviation multiple of the mean. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Finally, given the high prevalence of zero-application tract-years in both SBI (33.7 percent) and high-tech (85.1 percent), decomposing applications per capita into extensive and intensive margins provides additional insight into which component is most salient——and how this differs across sectors. In many cases, the same component of applications per capita drives the overall relationship in both sectors. For example, the negative association between median age and applications per capita appears to be driven by the extensive margin in both SBI and high-tech (the intensive margin is positive), while the negative relationship with average firm size is primarily driven by the intensive margin.

At the same time, there are cases where the dominant component of applications per capita differs across sectors. For example, although Hispanic share is negatively related to applications per capita in both SBI and high-tech, the source of this relationship differs: in SBI, the intensive margin plays the larger role, whereas in high-tech, the extensive margin is more important.

Having examined how individual local conditions relate to startups per capita, applica-

tions per capita and transition rates in SBI and high-tech industries, we next ask whether the contribution of broader groups of covariates also differs across sectors. To do so, we apply the regression decomposition used in the baseline analysis to each industry separately. The resulting industry-specific decompositions (Tables 15 and 16) reveal patterns broadly consistent with the all-industry results, but with notable differences across sectors.

Table 15: Small business intensive regression decomposition

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
Groups					
Demographic	0.024	0.002	-0.000	0.029	0.018
HH economic conditions	0.019	0.032	0.027	0.009	0.004
Incumbent firm characteristics	-0.002	0.001	0.011	-0.020	0.002
Commercial share	0.093	0.116	0.024	0.250	0.003
Categories					
Local conditions	0.134	0.151	0.062	0.268	0.028
Common market conditions	0.070	0.131	0.129	0.084	0.090
Residual	0.796	0.718	0.808	0.648	0.882

Notes: Decomposition of R^2 across variable groups for DHS(startups pc), DHS(apps pc), apps pc extensive, DHS(apps pc intensive), and transition rates for the SBI sample."DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

For SBI, the baseline pattern holds: among local conditions, household economic conditions and demographic characteristics explain the most variation in startups per capita. Household economic conditions account for more of the variation in applications per capita, whereas demographic characteristics play a relatively stronger role in transition rates. In high-tech, by contrast, demographic characteristics dominate, explaining most of the variation attributed to local conditions for startups per capita and for both margins. Finally, consistent with the baseline, in both sectors local conditions explain substantially more of the variation in applications per capita than in transition rates or startups per capita.

With county–year fixed effects included, the strong role of demographic characteristics in explaining high-tech startup activity reflects meaningful within-county heterogeneity rather than cross-metro differences. Demographic composition may proxy for highly localized concentrations of STEM human capital, immigrant professional networks, and knowledge spillovers that shape innovation-driven entrepreneurship. The positive associations between high-tech applications and both Asian-American and African-American population shares

Table 16: High-tech regression decomposition

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
Groups					
Demographic	0.013	0.035	0.034	0.054	0.007
HH economic conditions	0.003	0.021	0.021	0.001	0.001
Incumbent firm characteristics	0.004	0.008	0.007	-0.009	0.003
Commercial share	0.027	0.027	0.020	0.203	0.001
Categories					
Local conditions	0.048	0.090	0.082	0.249	0.011
Common market conditions	0.089	0.102	0.102	0.105	0.087
Residual	0.863	0.808	0.816	0.646	0.902

Notes: Decomposition of R^2 across variable groups for DHS(startups pc), DHS(apps pc), apps pc extensive, DHS(apps pc intensive), and transition rates for the high-tech sample."DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

therefore appear to capture tract-level sorting and micro-spatial ecosystem features, rather than broader regional demographic pattern.

8 Robustness and extensions

We supplement our core results with two robustness exercises that assess whether our findings are robust to accounting for intermediate spatial spillovers and compositional differences in ideas. First, because own-tract characteristics and county-by-year fixed effects may not fully capture the consumer, product, and labor market environments nascent entrepreneurs face, we extend our regressions to include characteristics of neighboring tracts, allowing us to account for intermediate spatial spillovers. Second, because observable local conditions may partially reflect differences in the composition or quality of ideas across locations, we conduct an application-level analysis that controls for detailed application characteristics.

Because both exercises support our main results, we provide a brief summary here and report additional details in the appendix. First, the relationships between own-tract characteristics and startups per capita, applications per capita, and transition rates are remarkably robust to the inclusion of neighboring tract characteristics (see Appendix E.1). Although neighboring tract characteristics are themselves informative, their inclusion does not alter the estimated relationships for own-tract conditions. For example, the offsetting relationship be-

tween African American population share and applications per capita versus transition rates are largely driven by own-tract characteristics.

Second, application-level analysis of transition rates confirms the tract-level patterns (see Appendix E.2). This exercise addresses the concern that tract-level results may reflect unobserved differences in the composition or quality of ideas across applications. The application-level data allow us to control directly for detailed industry classifications and other characteristics—such as legal form and reason for applying—that can be viewed as proxies for idea quality. Consistent with tract-level results, local conditions have similar relationships – e.g., locations with higher African American share have lower transition probabilities, and locations with higher house values have higher transition probabilities, even after conditioning on application characteristics. These patterns hold across all three industry samples. In short, our transition results on the role of local conditions are robust to these additional controls for application characteristics.

9 Conclusion

Startups expand economic opportunity, benefiting entrepreneurs, the workers they employ, and the communities in which they operate. Yet how and why some regions in the U.S. exhibit vibrant entrepreneurship while others are effectively startup deserts remains understudied. This paper sheds light on spatial variation in nascent entrepreneurship using novel data and a two-stage model of entry. Based on this framework, local employer startup rates are decomposed into two distinct components: idea generation, proxied by business applications per capita, and the rate at which those ideas transition into employer startups.

We find that both application intensity and transition rates contribute meaningfully to spatial variation in startups per capita across U.S. census tracts. Application intensity accounts for approximately two-thirds of this variation, while transition rates account for roughly one-third. Moreover, high-startup tracts are characterized by high application rates, whereas low-startup areas are marked by low transition rates.

The relationship between startup activity and local conditions reflects distinct associations with applications and transition rates. Some local characteristics—such as the shares of African-American and foreign-born residents—are positively related to application rates but negatively related to transition rates. In these cases, the negative relationship with transitions dampens, or even reverses, the overall relationship with startup rates. The share of African-American residents provides a striking example: it is positively associated with applications, but negatively associated with both transition rates and startups per capita.

Recognizing sectoral differences in entrepreneurial motivations and barriers, we extend

our analysis to two broad industry groups: small business–intensive industries, where local-market orientation and non-pecuniary motivations such as autonomy may matter more, and high-tech industries, often associated with innovation and growth. Spatial patterns are broadly similar across the two groups, but important quantitative differences emerge. High-tech entrepreneurship is more geographically concentrated, and application and transition rates are strongly negatively correlated, suggesting that places conducive to high-tech idea generation may not equally support successful transitions. The local share of highly educated individuals is especially predictive of high-tech applications, though observable characteristics are less helpful in identifying "superstar" high-tech locations.

Overall, our findings indicate that employer entry is a multi-stage process with a distinct nascent phase over an extended period of time. Although the analysis is descriptive and does not identify causal relationships, the results demonstrate that the determinants of entrepreneurial activity differ systematically between the idea generation stage and the transition from application to firm formation. This distinction is central to interpreting spatial variation in employer startup activity. In regions with low application rates, constraints may primarily reflect a limited supply of entrepreneurial ideas. In contrast, in areas characterized by low transition rates, frictions are more likely to arise at later stages of the entrepreneurial process, after ideas have already been formed. Identifying which margin is more binding in a given location can help analysts, policymakers, and local stakeholders better diagnose the sources of entrepreneurial disparities—whether they stem from differential exposure to entrepreneurial opportunities, variation in access to complementary inputs required for business formation, or other localized barriers that impede progression from intention to realization.

Given the distinct roles of the idea generation and transition stages in the formation of new employer firms, elucidating the mechanisms underlying these differences represents an important avenue for future research. In particular, a natural direction for further inquiry is the role of financial frictions in explaining why certain demographic groups, such as African Americans and the foreign-born, exhibit relatively high application intensity yet comparatively low transition rates.

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Appendix (Not Intended For Publication)

A Additional tables

Table A.1 defines each variable included in our regression analysis.

Table A.1: Description of local condition variables

Variable	**Definition**	**Source**
log(median age)	log of median age	ACS
BA or higher share	share of pop. with BA or higher degree	ACS
some college share	share of pop. with some college	ACS
African American share	African American pop. share	ACS
Asian share	Asian pop.share	ACS
Hispanic share	Hispanic pop. share	ACS
foreign born share	foreign both pop. share	ACS
log(per capita income)	log of per capita income	ACS
unemployment rate	unemployment rate	ACS
log(median house value)	log of the median house value	ACS
share of emp in young firms	share of emp. in firms aged 1-5	LBD
share of emp in large firms	share of emp. in firms with 500+ emp	LBD
DHS(average emp)	DHS of the average emp. of firms	LBD
Industry emp. shares	3-digit NAICS employment shares	LBD
commercial share	emp. share = emp./ (pop. + emp.)	ACS & LBD

Notes: "DHS" refers to the transformation based on Davis et al. (1996), where the deviation is taken from the grand mean.

B Business application results

Business applications (BA) contain all applications from the BFS, including wage (WBA) and non-wage applications.[41] Overall, the results using BA are qualitatively similar to the WBA results. In this appendix section, we provide a brief overview of these findings, focusing first on the basic data patterns and then moving on to the regression analysis.

The basic patterns observed for WBA—substantial variation in startups per capita, applications per capita, and transition rates across counties and tracts—are also observed for BA. Looking at the descriptive statistics for BA in Table B.1, startup per capita (defined as per 1,000 prime age adults) averaged 1.01 and 1.56, at the county and tract levels, respectively. This reflects applications per capita of 8.15 and 13.14 with transition rates of 0.12 and 0.12, for the county and tract data. Compared with the WBA statistics in Table 1, application activity is much higher, as this reflects the large number of business applications that do not report a planned wage date (non-WBA cases). These non-WBA cases have low transition rates compared to WBA cases, and hence the overall BA sample has transition rates that are much lower than those for WBA. What is quite similar across BA and WBA, however, are the coefficients of variation for all three measures at the county and tract levels. The BA data show considerable variability across locations, especially at the tract level.

The variance decomposition results for startups per capita from BA presented in Table B.2 mimic those of the variance decomposition reported in Table 2 for WBA. The majority of the variance in startups per capita is accounted for by the variance in applications per capita and the covariance term accounts for a very small share of the overall variance.

[41] The one difference relative to the published BFS is that we omit applications submitted due to changes in organizational form or the purchase of an existing business–to focus on cases more closely associated with entirely new business formation.

Table B.1: BA startups, applications, and transition rates: summary statistics

	County			Tract		
	(1)	(2)	(3)	(4)	(5)	(6)
	Startups pc	Applications pc	Transition rate	Startups pc	Applications pc	Transition rate
Mean	1.012	8.154	0.122	1.564	13.140	0.116
SD	0.562	4.686	0.037	8.405	50.320	0.054
CV	0.555	0.575	0.307	5.374	3.830	0.464

Notes: Mean, standard deviation (SD), and coefficient of variation (CV) of BA startups per 1,000 prime-age people (startups pc), BA per capita (applications pc), and BA transition rate (startups divided by applications) between 2011 and 2016, reported separately for county- and tract-level data.

Table B.2: BA variance decomposition

	County			Tract		
	(1)	(2)	(3)	(4)	(5)	(6)
	Applications pc	Transition rate	2 × covariance	Applications pc	Transition rate	2 × covariance
BA	0.670	0.290	0.040	0.579	0.424	-0.002

Notes: Variance decomposition of log(BA startups pc) into log(BA pc) and log(BA transition rate) for 2011-2016, reported separately for county- and tract-level data. "pc" refers to per capita (per 1,000 prime-age people).

Table B.3 reports the regression results for BA. The findings are very similar to those reported for WBA in Table 5. There are, however, a few notable differences. Application activity is positively associated with the bachelors or higher degree share whereas there is no statistically significant association in the WBA regressions. Transition rates are negatively related to foreign born share in the case of WBA but there is little association in the case of BA. Finally, the overall and within R^2 values are higher in the BA models compared to the WBA models. Local attributes explain more of the overall variation in startups per capita, applications per capita, and transition rates and more of the within variation, as well.

Table B.3: BA regression results

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Transition rate
log(median age)	0.0947***	0.167***	0.0110***
	(0.029)	(0.0269)	(0.00251)
bachelors or higher share	0.244***	0.210***	0.00143
	(0.0459)	(0.0381)	(0.00426)
some college share	–0.207***	–0.0496	–0.0275***
	(0.0523)	(0.0484)	(0.00424)
African American share	–0.230***	0.601***	–0.0637***
	(0.062)	(0.061)	(0.00343)
Asian share	0.102	–0.169*	0.0544***
	(0.147)	(0.0896)	(0.0142)
Hispanic share	–0.198**	–0.137	–0.00242
	(0.0832)	(0.0964)	(0.00601)
foreign born share	0.515***	0.446***	–0.00759
	(0.124)	(0.14)	(0.00681)
log(per capita income)	0.205***	0.245***	–0.00827***
	(0.027)	(0.0244)	(0.00174)
unemployment rate	0.419***	0.574***	–0.00326
	(0.0574)	(0.0481)	(0.00553)
log(median house value)	0.252***	0.107***	0.0159***
	(0.0182)	(0.0137)	(0.000922)
share of emp in young firms	0.223***	0.145***	0.00772***
	(0.0244)	(0.0156)	(0.00224)
share of emp in large firms	–0.567***	–0.380***	–0.0268***
	(0.0218)	(0.0141)	(0.00165)
DHS(avg firm emp)	–0.171***	–0.131***	–0.00814***
	(0.00604)	(0.00432)	(0.000482)
commercial share	2.617***	1.814***	0.117***
	(0.032)	(0.0343)	(0.00284)
Obs	428,000	430,000	428,000
R-squared	0.4356	0.6878	0.2146
Within R-squared	0.2984	0.4328	0.0769

Notes: Regressions use tract-level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. The observation counts have been rounded for disclosure reasons. ***, **, and * indicate sig. at the 1%, 5%, and 10% levels. Standard errors are clustered at the county level.

C Baseline analysis

C.1 Augmented results

Because our baseline analysis pools across all industries, only 7.4 percent of tract-years have no applications, making the distinction between the extensive and intensive margins

of applications per capita less salient for the baseline analysis. Nevertheless, for robustness, we report regression results for two additional outcomes in Tables C.1, C.2 and C.3: an indicator for positive applications (extensive margin) and positive applications per capita (intensive margin).

Table C.1: Augmented baseline regression results

	(1)	(2)	(3)	(4)	(5)
	DHS(startups pc)	DHS(applications pc)	Apps pc extensive	DHS(apps pc intensive)	Transition rate
log(median age)	0.138***	0.126***	–0.0402***	0.226***	0.0301***
	(0.0261)	(0.0247)	(0.00866)	(0.0224)	(0.0056)
bachelors or higher share	0.179***	0.0447	–0.0477***	0.101***	0.0506***
	(0.0408)	(0.037)	(0.0126)	(0.0362)	(0.0104)
some college share	–0.308***	–0.161***	0.0480***	–0.260***	–0.0513***
	(0.0522)	(0.05)	(0.0143)	(0.0439)	(0.0116)
African American share	–0.267***	0.522***	0.0620***	0.469***	–0.197***
	(0.0378)	(0.0402)	(0.00768)	(0.0361)	(0.00654)
Asian share	0.0729	0.079	0.0540***	0.00747	0.0182
	(0.109)	(0.0924)	(0.0202)	(0.0813)	(0.0228)
Hispanic share	–0.181**	–0.0881	0.0459**	–0.142	–0.018
	(0.0784)	(0.107)	(0.0187)	(0.0886)	(0.0138)
foreign born share	0.271**	0.387***	0.0753***	0.285**	–0.0536***
	(0.109)	(0.142)	(0.0231)	(0.118)	(0.0141)
log(per capita income)	0.152***	0.231***	0.0720***	0.140***	–0.0152***
	(0.0184)	(0.02)	(0.00575)	(0.0202)	(0.00416)
unemployment rate	0.447***	0.563***	–0.0289*	0.698***	–0.0214
	(0.0589)	(0.0513)	(0.017)	(0.0529)	(0.0145)
log(median house value)	0.213***	0.142***	0.0293***	0.0999***	0.0385***
	(0.0156)	(0.0154)	(0.00456)	(0.0129)	(0.00231)
share of emp in young firms	0.210***	0.176***	0.00803	0.181***	0.00455
	(0.027)	(0.0197)	(0.00742)	(0.0176)	(0.00648)
share of emp in large firms	–0.534***	–0.557***	–0.0598***	–0.477***	–0.0217***
	(0.0238)	(0.0183)	(0.00554)	(0.0165)	(0.0047)
DHS(avg firm emp)	–0.183***	–0.148***	0.00916***	–0.178***	–0.0108***
	(0.00588)	(0.00555)	(0.00195)	(0.00488)	(0.00142)
commercial share	2.694***	2.458***	0.126***	2.369***	0.144***
	(0.0343)	(0.0336)	(0.00744)	(0.0322)	(0.00538)
Obs	398,000	430,000	430,000	398,000	398,000
R-squared	0.3079	0.4203	0.1756	0.4402	0.1453
Within R-squared	0.2218	0.2677	0.04166	0.3171	0.04173

Notes: Regressions use tract-level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. Standard errors clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.2: Augmented baseline regression elasticities

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
log(median age)	0.138	0.126	-0.043	0.226	0.081
bachelors or higher share	0.051	0.012	-0.014	0.029	0.039
some college share	-0.088	-0.046	0.015	-0.075	-0.040
African American share	-0.037	0.072	0.009	0.064	-0.072
Asian share	0.003	0.004	0.003	0.000	0.002
Hispanic share	-0.028	-0.014	0.008	-0.022	-0.007
foreign born share	0.034	0.047	0.010	0.036	-0.018
log(per capita income)	0.152	0.231	0.078	0.140	-0.041
unemployment rate	0.041	0.053	-0.003	0.065	-0.005
log(median house value)	0.213	0.142	0.032	0.100	0.103
share of emp in young firms	0.037	0.031	0.002	0.032	0.002
share of emp in large firms	-0.034	-0.035	-0.004	-0.031	-0.004
DHS(avg firm emp)	-0.183	-0.148	0.010	-0.178	-0.029
commercial share	0.501	0.441	0.024	0.440	0.072

Notes: Estimated percent change in the LHS variable induced by a one percent change in the RHS variable. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Table C.3: Augmented baseline regression magnitudes

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(appc pc intensive)	(5) Transition rate
log(median age)	2.663	2.470	-0.843	4.362	1.563
bachelors or higher share	3.325	0.798	-0.931	1.891	2.543
some college share	-2.411	-1.274	0.416	-2.055	-1.069
African American share	-5.994	11.722	1.465	10.368	-11.664
Asian share	0.566	0.772	0.579	0.000	0.378
Hispanic share	-3.800	-1.789	1.101	-2.985	-0.950
foreign born share	3.703	5.347	1.114	3.920	-1.960
log(per capita income)	7.843	11.966	4.040	7.224	-2.116
unemployment rate	2.591	3.313	-0.187	4.010	-0.308
log(median house value)	16.891	11.445	2.579	7.930	8.168
share of emp in young firms	2.512	2.185	0.141	2.173	0.136
share of emp in large firms	-8.452	-8.908	-1.018	-7.707	-0.994
DHS(avg firm emp)	-24.614	-21.712	1.467	-23.941	-3.901
commercial share	46.192	41.624	2.270	40.660	6.638

Notes: Estimated percent change in the LHS variable induced by the percent change in the RHS variable equivalent to a one standard deviation multiple of the mean. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

C.2 Testing equality across elasticities

Table C.4 reports p-values from pairwise tests of equality across elasticities estimated in a seemingly unrelated regression (SUR) framework for our three outcomes—startups per capita, applications per capita, and transition rates. The SUR is estimated on the sample of tract-years with positive applications, which provides common support for startups and transition rates. The reported p-values correspond to tests of whether differences in elasticities across outcomes for a given covariate are statistically different from zero.

Table C.4: p-value table: all industries

	Difference in Elasticity between:		
	(1)	(2)	(3)
	Startups pc & Apps pc	Startups pc & Transiton rates	Apps pc & Transition rates
log(median age)	0.000	0.009	0.000
bachelors or higher share	0.004	0.156	0.426
some college share	0.121	0.000	0.029
African American share	0.000	0.000	0.000
Asian share	0.268	0.770	0.646
Hispanic share	0.237	0.167	0.412
foreign born share	0.760	0.001	0.002
log(per capita income)	0.335	0.000	0.000
unemployment rate	0.000	0.000	0.000
log(median house value)	0.000	0.000	0.797
share of emp in young firms	0.143	0.000	0.000
share of emp in large firms	0.000	0.000	0.000
DHS(avg firm emp)	0.179	0.000	0.000
commercial share	0.000	0.000	0.000

Notes: p-values are based on seemingly unrelated regression estimates of the three outcomes using the sample of tract-years with positive applications. The tests evaluate whether differences in elasticities across outcomes for a given covariate are statistically different from zero..

D Industry analysis

D.1 Additional Tables

Table D.1 reports the tract-level WBA variance decomposition for three samples: all-industry (baseline), SBI, and high-tech. Table D.2 provides summary statistics for outcome variables and local covariates separately for the SBI and high-tech samples. Table D.3 reports the contribution of county-by-year fixed effects to startups per capita, applications per capita, and transition rates by industry. Tables D.4 through D.9 present the full set of regression results for the SBI and high-tech samples.

Table D.1: Variance decomposition for wage startups per capita (industry samples)

	(1) Applications pc	(2) Transition rate	(3) 2 × covariance
All Tracts	0.659	0.325	0.016
Small Business Intensive Tracts	0.721	0.287	-0.008
High-Tech Tracts	0.825	0.182	-0.007

Notes: Variance decomposition of log(WBA startups pc) into log(WBA pc) and log(WBA transition rate) for 2011-2016 for the all (row 1), small business intensive (row 2), and high-tech (row 3) industry samples. "pc" denotes per capita (per 1,000 prime-age people).

Table D.2: Regression variable summary statistics by industry

	Small Business Intensive			High-Tech		
	(1) Mean	(2) SD	(3) CV	(4) Mean	(5) SD	(6) CV
DHS(startups pc)	-0.663	1.152	-1.738	-0.843	1.287	-1.526
DHS(apps pc)	-0.621	1.102	-1.776	-1.491	1.191	-0.799
apps pc extensive	0.668	0.471	0.705	0.156	0.363	2.329
DHS(apps pc intensive)	-0.294	0.632	-2.150	-0.251	0.546	-2.175
transition rate	0.402	0.389	0.968	0.386	0.450	1.166
log(median age)	3.627	0.192	0.053	3.632	0.185	0.051
bachelors or higher share	0.308	0.191	0.618	0.427	0.202	0.474
some college share	0.286	0.079	0.277	0.269	0.084	0.313
African American share	0.130	0.211	1.621	0.098	0.160	1.623
Asian share	0.052	0.091	1.763	0.086	0.117	1.366
Hispanic share	0.158	0.206	1.305	0.141	0.167	1.188
foreign born share	0.134	0.137	1.023	0.164	0.137	0.833
log(per capita income)	10.200	0.459	0.045	10.440	0.467	0.045
unemployment rate	0.090	0.055	0.609	0.075	0.044	0.586
log(median house value)	12.130	1.176	0.097	12.440	1.301	0.105
share of emp in young firms	0.176	0.114	0.649	0.186	0.114	0.616
share of emp in large firms	0.069	0.162	2.332	0.091	0.175	1.920
DHS(avg firm emp)	-0.250	0.637	-2.545	-0.210	0.675	-3.213
commercial share	0.203	0.179	0.885	0.254	0.218	0.861

Notes: Mean, standard deviation (SD), and coefficient of variation (CV) of variables used as dependent and control variables in small business intensive (cols. 1-3) and high-tech (cols. 5-6) regressions. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Table D.3: Contribution of location fixed effects by industry

	Small Business Intensive			High-Tech		
	(1)	(2)	(3)	(4)	(5)	(6)
	DHS(Startups pc)	DHS(Applications pc)	Transition rate	DHS(Startups pc)	DHS(Applications pc)	Transition rate
R-squared	0.081	0.154	0.093	0.094	0.113	0.088
Obs	285,000	430,0000	285,000	64,000	430,000	64,000

Notes: Share of variance in DHS(startups per capita), DHS(applications per capita), and transition rates explained by county-by-year fixed effects in the SBI (cols. 1-3) and high-tech (cols. 4-6) tract-year samples (2011-2016).

Table D.4: Small business intensive regression results

	(1)	(2)	(3)	(4)	(5)
	DHS(startups pc)	DHS(applications pc)	Apps pc extensive	DHS(apps pc intensive)	Transition rate
log(median age)	0.150***	–0.0339	–0.114***	0.359***	0.0251***
	(0.0302)	(0.0272)	(0.0128)	(0.0265)	(0.0074)
bachelors or higher share	0.171***	0.152***	0.0464**	0.0864**	0.0588***
	(0.0518)	(0.0457)	(0.0187)	(0.0412)	(0.0155)
some college share	–0.397***	–0.0252	0.203***	–0.551***	–0.0402**
	(0.0557)	(0.0517)	(0.0231)	(0.0372)	(0.017)
African American share	–0.500***	0.226***	0.0447***	0.241***	–0.223***
	(0.026)	(0.0356)	(0.0134)	(0.0252)	(0.00668)
Asian share	–0.324***	–0.223**	–0.00403	–0.262***	–0.0427*
	(0.0864)	(0.0938)	(0.0317)	(0.0644)	(0.0224)
Hispanic share	–0.233***	–0.147	–0.00302	–0.147**	–0.0386***
	(0.0713)	(0.115)	(0.0342)	(0.0712)	(0.0136)
foreign born share	0.191**	0.491***	0.229***	0.053	–0.0270*
	(0.0832)	(0.136)	(0.0397)	(0.0921)	(0.0143)
log(per capita income)	0.116***	0.277***	0.121***	0.0460*	–0.0117**
	(0.0237)	(0.0234)	(0.00786)	(0.0253)	(0.00559)
unemployment rate	0.391***	0.479***	–0.0261	0.868***	–0.0437**
	(0.0828)	(0.0702)	(0.0285)	(0.0649)	(0.0219)
log(median house value)	0.171***	0.169***	0.0664***	0.0493***	0.0408***
	(0.0172)	(0.0193)	(0.00775)	(0.014)	(0.0035)
share of emp in young firms	0.136***	0.146***	0.0369***	0.108***	–0.00127
	(0.0321)	(0.0211)	(0.0103)	(0.018)	(0.01)
share of emp in large firms	–0.383***	–0.582***	–0.141***	–0.377***	–0.0145*
	(0.0277)	(0.0205)	(0.00878)	(0.014)	(0.00747)
DHS(avg firm emp)	–0.195***	–0.129***	0.00669**	–0.211***	–0.0136***
	(0.00814)	(0.00722)	(0.00318)	(0.00565)	(0.00218)
commercial share	2.270***	2.443***	0.425***	2.121***	0.129***
	(0.0366)	(0.0383)	(0.0144)	(0.0307)	(0.0074)
Obs	285,000	430,000	430,000	285,000	285,000
R-squared	0.204	0.2821	0.1916	0.3518	0.1178
Within R-squared	0.1342	0.1512	0.06227	0.2678	0.02757

Notes: Regressions use tract-year–level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.5: Small business intensive regression elasticities

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
log(median age)	0.150	-0.034	-0.171	0.359	0.063
bachelors or higher share	0.053	0.042	0.019	0.027	0.045
some college share	-0.114	-0.007	0.087	-0.158	-0.029
African American share	-0.065	0.031	0.009	0.031	-0.072
Asian share	-0.017	-0.010	-0.000	-0.014	-0.005
Hispanic share	-0.037	-0.023	-0.001	-0.023	-0.015
foreign born share	0.026	0.060	0.042	0.007	-0.009
log(per capita income)	0.116	0.277	0.181	0.046	-0.029
unemployment rate	0.035	0.045	-0.004	0.078	-0.010
log(median house value)	0.171	0.169	0.099	0.049	0.102
share of emp in young firms	0.024	0.026	0.010	0.019	-0.001
share of emp in large firms	-0.026	-0.036	-0.013	-0.026	-0.002
DHS(avg firm emp)	-0.195	-0.129	0.010	-0.211	-0.034
commercial share	0.461	0.438	0.114	0.430	0.065

Notes: Estimated percent change in the LHS variable induced by a one percent change in the RHS variable. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Table D.6: p-value table: small business intensive

	Difference in Elasticity between:		
	(1) Startups pc & Apps pc	(2) Startups pc & Transiton rates	(3) Apps pc & Transition rates
log(median age)	0.000	0.002	0.000
bachelors or higher share	0.011	0.003	0.165
some college share	0.000	0.000	0.000
African American share	0.000	0.000	0.000
Asian share	0.001	0.007	0.808
Hispanic share	0.227	0.155	0.889
foreign born share	0.382	0.170	0.964
log(per capita income)	0.180	0.001	0.482
unemployment rate	0.000	0.000	0.000
log(median house value)	0.000	0.415	0.000
share of emp in young firms	0.308	0.001	0.786
share of emp in large firms	0.081	0.000	0.000
DHS(avg firm emp)	0.000	0.000	0.000
commercial share	0.000	0.000	0.000

Notes: See notes to Table C.4.

Table D.7: High-tech regression results

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
log(median age)	0.0918*	–0.348***	–0.116***	0.507***	0.000235
	(0.0532)	(0.0316)	(0.00945)	(0.044)	(0.0165)
bachelors or higher share	0.308***	0.905***	0.285***	0.058	0.0729**
	(0.0978)	(0.0557)	(0.017)	(0.052)	(0.0312)
some college share	–0.282**	–0.200***	–0.0226	–0.839***	0.0449
	(0.133)	(0.077)	(0.0222)	(0.0684)	(0.0426)
African American share	–0.306***	0.0684***	0.0171**	0.197***	–0.167***
	(0.0627)	(0.0255)	(0.0075)	(0.0397)	(0.0201)
Asian share	0.434***	0.596***	0.183***	0.147*	0.0656**
	(0.146)	(0.218)	(0.0632)	(0.0872)	(0.0322)
Hispanic share	–0.176*	–0.297***	–0.0876***	–0.0957	–0.0411
	(0.0951)	(0.0701)	(0.0216)	(0.0641)	(0.0284)
foreign born share	–0.119	0.439***	0.138***	–0.194***	–0.0765**
	(0.0954)	(0.103)	(0.0315)	(0.0603)	(0.0311)
log(per capita income)	0.0399	0.265***	0.0777***	–0.00805	–0.0142
	(0.0361)	(0.0203)	(0.00569)	(0.0298)	(0.0105)
unemployment rate	0.273	0.262***	0.0609***	0.940***	–0.0592
	(0.166)	(0.0562)	(0.0169)	(0.0914)	(0.0545)
log(median house value)	0.0631***	0.0505***	0.0165***	–0.0208	0.0280***
	(0.0203)	(0.013)	(0.004)	(0.0181)	(0.00664)
share of emp in young firms	0.0989	0.190***	0.0590***	0.0253	0.00206
	(0.0684)	(0.0199)	(0.00626)	(0.0363)	(0.0236)
share of emp in large firms	–0.128**	–0.193***	–0.0492***	–0.208***	0.00201
	(0.0497)	(0.0246)	(0.00737)	(0.0252)	(0.0162)
DHS(avg firm emp)	–0.145***	–0.0318***	–0.00312	–0.213***	–0.0115**
	(0.0158)	(0.00671)	(0.00207)	(0.011)	(0.00501)
commercial share	1.108***	1.119***	0.285***	1.412***	0.0464***
	(0.061)	(0.0548)	(0.0154)	(0.0485)	(0.0145)
Obs	64,000	430,000	430,000	64,000	64,000
R-squared	0.1368	0.1924	0.1845	0.3544	0.09781
Within R-squared	0.04759	0.08992	0.08225	0.2489	0.01128

Notes: Regressions use tract-year–level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.8: High-tech regression elasticities

	(1) DHS(startups pc)	(2) DHS(applications pc)	(3) Apps pc extensive	(4) DHS(apps pc intensive)	(5) Transition rate
log(median age)	0.092	-0.348	-0.745	0.507	0.001
bachelors or higher share	0.132	0.251	0.508	0.025	0.081
some college share	-0.076	-0.057	-0.042	-0.225	0.031
African American share	-0.030	0.009	0.015	0.019	-0.043
Asian share	0.037	0.027	0.053	0.013	0.014
Hispanic share	-0.025	-0.045	-0.086	-0.014	-0.015
foreign born share	-0.020	0.054	0.109	-0.032	-0.032
log(per capita income)	0.040	0.265	0.499	-0.008	-0.037
unemployment rate	0.021	0.025	0.037	0.071	-0.012
log(median house value)	0.063	0.051	0.106	-0.021	0.073
share of emp in young firms	0.018	0.033	0.067	0.005	0.001
share of emp in large firms	-0.012	-0.012	-0.020	-0.019	0.001
DHS(avg firm emp)	-0.145	-0.032	-0.020	-0.213	-0.030
commercial share	0.281	0.201	0.329	0.358	0.031

Notes: Estimated percent change in the LHS variable induced by a one percent change in the RHS variable. The LHS variable of the regression is listed in the columns, and each RHS variable is listed in the rows. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

Table D.9: p-value table: high-tech

	Difference in Elasticity between:		
	(1) Startups pc & Apps pc	(2) Startups pc & Transiton rates	(3) Apps pc & Transition rates
log(median age)	0.000	0.000	0.000
bachelors or higher share	0.044	0.471	0.273
some college share	0.001	0.000	0.000
African American share	0.000	0.001	0.000
Asian share	0.266	0.003	0.074
Hispanic share	0.004	0.090	0.569
foreign born share	0.001	0.007	0.298
log(per capita income)	0.000	0.000	0.019
unemployment rate	0.000	0.000	0.000
log(median house value)	0.000	0.000	0.001
share of emp in young firms	0.340	0.000	0.000
share of emp in large firms	0.771	0.000	0.000
DHS(avg firm emp)	0.018	0.000	0.000
commercial share	0.000	0.000	0.000

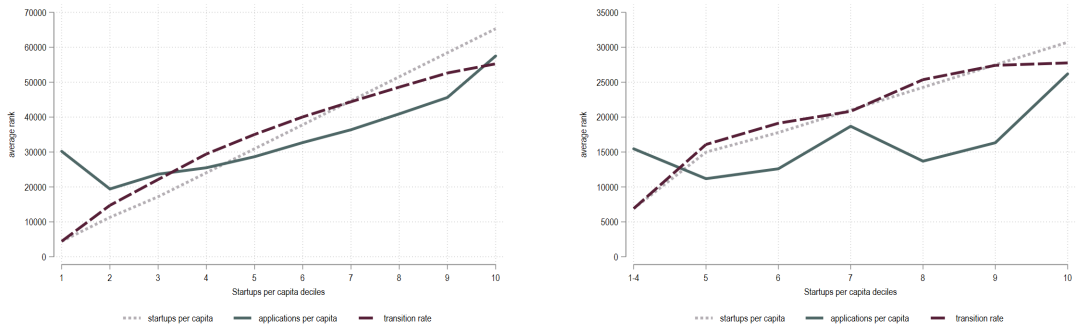
Notes: See notes to Table C.4.

D.2 Ranking analysis

Basic ranking analysis: Figure D.1 depicts the SBI and high-tech counterparts to Figure 4. This ranking analysis should be interpreted with caution, especially for high-tech, where many tracts have zero applications and startups. As a result, the smaller number of active tracts and deciles produces lumpier patterns.

For SBI, the patterns closely mirror the all-industry results: high-startup tracts are characterized by high applications per capita, while low-startup tracts are marked by low transition rates.[42] The high-tech patterns differ in two important ways. First, a large share of tracts record no high-tech applications, reflecting the sector's strong spatial concentration. Even among tracts with activity, the lowest startup-intensity deciles cluster tightly because many tracts have zero high-tech startups. Second, relative to SBI and the pooled sample, applications per capita and transition rates contribute more similarly to variation in high-tech startup intensity, rather than one margin clearly dominating. At the lower end of the distribution, however, high-tech resembles SBI and the all-industry samples: differences in startup intensity are more closely associated with transition rates than with applications per capita, consistent with the importance of implementation frictions in low-activity locations.

Figure D.1: Tract-level rank analysis by industry



(a) Small Business Intensive (b) High-tech

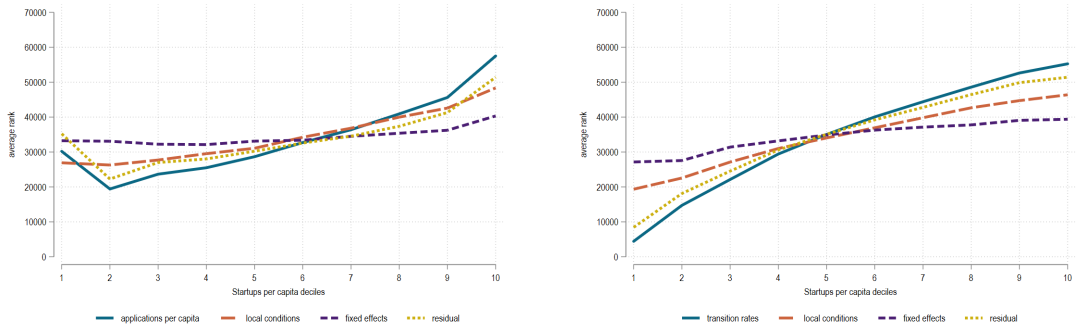
Notes: Average ranks for deciles for WBA startups per capita, applications per capita, and transition rates between 2011 and 2016 using tract-level data. Figure (a) is for the small business intensive sample and figure (b) is for the high-tech sample.

Ranking analysis revisited: Figures D.2 and D.3 are the SBI and high-tech coun-

[42] The dip in the first decile reflects tracts with zero startups per capita, some of which still report positive application activity.

terparts to Figure 5, respectively. For reasons discussed earlier, these results should be interpreted with caution. For SBI, the results broadly mirror, though are somewhat weaker than, the all-industry baseline. Observable characteristics track the raw data relatively well at the top of the startups per capita distribution for both applications per capita and transition rates, but are far less informative at the bottom of the distribution. By contrast, in high-tech, observable characteristics are much less informative, especially at the top of the startups per capita distribution.

Figure D.2: Tract-level small business intensive rank analysis based on estimates



(a) SBI Applications per capita

(b) SBI Transition rate

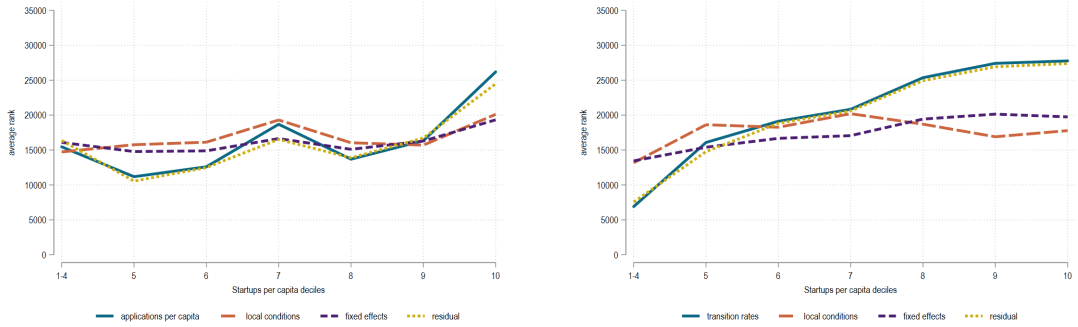
Notes: Rank analysis based on small business intensive WBA startups per capita at the tract level. Plots depict average ranks by decile using observed SBI applications per capita (left panel) and transition rates (right panel), along with corresponding predicted values based on local conditions, county-by-year fixed effects and the residual.

E Robustness

E.1 Controlling for neighboring tract characteristics

Own-tract characteristics and county-by-year fixed effects may not fully capture the consumer, product, and labor market conditions that nascent entrepreneurs face. The former reflect very localized conditions, while the latter aggregate over much broader areas, potentially missing patterns between these two geographic units. To address this, we re-estimate our models with an expanded set of controls that include characteristics of neighboring tracts. Specifically, we augment the baseline specification by adding the

Figure D.3: Tract-level high-tech rank analysis based on estimates



(a) High-tech applications per capita (b) High-tech transition rate

Notes: Rank analysis based on high-tech WBA startups per capita at the tract level. Plots depict average ranks by decile using observed high-tech applications per capita (left panel) and transition rates (right panel), along with the corresponding predicted values based on local conditions, county-fixed effects and the residual.

same set of variables—demographics, household economic conditions, and incumbent firm characteristics—measured as the mean across adjacent tracts.

Table E.1 reports results for the all-industry sample, with separate columns for own-tract and neighboring-tract coefficients. Table E.2 presents analogous results for the SBI and high-tech samples, where only the own-tract coefficients are reported.

Across all specifications, the own-tract results remain highly stable. In the all-industry sample, across all three dependent variables, the signs and significance of own-tract coefficients are virtually unchanged when neighbor controls are included.[43] In the industry-specific models, some coefficients vary modestly in significance, but the magnitudes and overall patterns remain similar.

The contribution of neighboring-tract characteristics to explanatory power is modest. As shown in Table E.3, adding neighbor controls slightly reduces the explanatory share of own-tract variables and marginally increases the overall R^2, but does not materially improve the model's ability to explain within-county variation. Taken together, these results support the view that startups per capita, applications per capita, and transition rates are heavily influenced by highly local, tract-specific conditions, with neighboring tracts playing a secondary, complementary role.

[43] A few statistically insignificant covariates exhibit sign changes, but these are rare exceptions.

Table E.1: Neighboring tract controls: all industries

	DHS(startups pc)		DHS(applications pc)		Transition rate	
	(1)	(2)	(3)	(4)	(5)	(6)
	own	neighbr	own	neighbor	own	neighbor
log(median age)	0.112***	0.0925*	0.107***	0.0422	0.0231***	0.0381***
	(0.0192)	(0.0497)	(0.018)	(0.0453)	(0.00449)	(0.00936)
bachelors or higher share	0.203***	–0.202***	0.000399	–0.0286	0.0698***	–0.0266
	(0.0333)	(0.0714)	(0.0278)	(0.0653)	(0.00934)	(0.0182)
some college share	–0.266***	–0.116	–0.174***	0.143	–0.0421***	–0.0546***
	(0.0375)	(0.0865)	(0.0333)	(0.088)	(0.0108)	(0.0204)
African American share	–0.273***	0.0478	0.306***	0.307***	–0.154***	–0.0480***
	(0.0295)	(0.0433)	(0.0239)	(0.0552)	(0.00649)	(0.00887)
Asian share	0.0456	0.108	0.110**	–0.00444	–0.0129	0.0566
	(0.0574)	(0.147)	(0.0486)	(0.156)	(0.0146)	(0.0381)
Hispanic share	–0.0821**	–0.0772	0.0221	–0.106	–0.00828	–0.00842
	(0.0412)	(0.0804)	(0.0415)	(0.107)	(0.00915)	(0.018)
foreign born share	0.162***	0.147	0.185***	0.366*	–0.0207*	–0.0841***
	(0.0559)	(0.151)	(0.0519)	(0.214)	(0.0115)	(0.0267)
log(per capita income)	0.136***	0.0552	0.216***	0.0646	–0.0109***	–0.0289***
	(0.0149)	(0.0416)	(0.0143)	(0.042)	(0.00386)	(0.0072)
unemployment rate	0.402***	0.500***	0.458***	0.884***	0.00164	–0.102***
	(0.0501)	(0.124)	(0.0396)	(0.122)	(0.0134)	(0.0284)
log(median house value)	0.165***	0.133***	0.116***	0.0881***	0.0290***	0.0286***
	(0.0124)	(0.0293)	(0.0136)	(0.0214)	(0.00208)	(0.00525)
share of emp in young firms	0.187***	0.338***	0.154***	0.325***	0.00362	0.00753
	(0.024)	(0.0526)	(0.0179)	(0.051)	(0.00611)	(0.013)
share of emp in large firms	–0.531***	0.0247	–0.548***	–0.0718*	–0.0229***	0.0268**
	(0.0231)	(0.0471)	(0.0168)	(0.0387)	(0.00467)	(0.0113)
DHS(avg firm emp)	–0.171***	–0.0503***	–0.140***	–0.0412***	–0.00932***	–0.00559***
	(0.00561)	(0.00862)	(0.005)	(0.00845)	(0.0014)	(0.00201)
commercial share	2.660***	0.054	2.412***	0.137***	0.150***	–0.0260***
	(0.0317)	(0.0424)	(0.0288)	(0.0416)	(0.00527)	(0.00913)
Obs	398,000		430,000		398,000	
R-squared	0.3106		0.425		0.1466	
Within R-squared	0.2248		0.2736		0.04327	

Notes: Regressions use tract-year-level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.2: Neighboring tract controls: small business intensive & high-tech

	Small Business Intensive			High-Tech		
	(1)	(2)	(3)	(4)	(5)	(6)
	DHS(startups pc)	DHS(applications pc)	Transition rate	DHS(startups pc)	DHS(applications pc)	Transition rate
log(median age)	0.131**	–0.264***	0.00618	0.135***	–0.0314	0.0184***
	(0.0526)	(0.0202)	(0.0173)	(0.0229)	(0.0211)	(0.00687)
bachelors or higher share	0.340***	0.633***	0.0985***	0.207***	0.0939**	0.0815***
	(0.102)	(0.0457)	(0.036)	(0.0438)	(0.0365)	(0.0145)
some college share	–0.0935	–0.0628	0.0519	–0.324***	–0.111***	–0.0386**
	(0.138)	(0.0475)	(0.0478)	(0.0485)	(0.0399)	(0.0158)
African American share	–0.136*	0.101***	–0.0895***	–0.406***	0.113***	–0.168***
	(0.0776)	(0.0268)	(0.0281)	(0.0306)	(0.0291)	(0.0099)
Asian share	0.477***	0.839***	0.0694	–0.248***	–0.0849	–0.0748***
	(0.175)	(0.139)	(0.0465)	(0.0784)	(0.0688)	(0.0204)
Hispanic share	–0.0327	–0.119***	–0.00377	–0.129***	0.0106	–0.0320**
	(0.127)	(0.0377)	(0.039)	(0.0466)	(0.0427)	(0.013)
foreign born share	–0.176	0.281***	–0.0862**	0.0979	0.238***	0.00594
	(0.111)	(0.0525)	(0.0414)	(0.0655)	(0.0535)	(0.0167)
log(per capita income)	0.019	0.219***	–0.0202*	0.106***	0.243***	–0.00563
	(0.0375)	(0.0163)	(0.0114)	(0.0193)	(0.0165)	(0.00542)
unemployment rate	0.204	0.172***	–0.0475	0.359***	0.405***	–0.008
	(0.159)	(0.0458)	(0.0517)	(0.0753)	(0.0572)	(0.0224)
log(median house value)	0.0602***	0.0528***	0.0233***	0.125***	0.132***	0.0297***
	(0.02)	(0.0109)	(0.00701)	(0.0137)	(0.0171)	(0.00331)
share of emp in young firms	0.0972	0.178***	0.000489	0.124***	0.131***	–0.00292
	(0.0695)	(0.0187)	(0.0244)	(0.0302)	(0.0205)	(0.00962)
share of emp in large firms	–0.123**	–0.187***	0.00208	–0.374***	–0.572***	–0.0146*
	(0.0511)	(0.0233)	(0.0171)	(0.0273)	(0.0189)	(0.00746)
DHS(avg firm emp)	–0.146***	–0.0231***	–0.0121**	–0.186***	–0.120***	–0.0130***
	(0.0159)	(0.00604)	(0.00522)	(0.0079)	(0.00665)	(0.00223)
commercial share	1.058***	1.028***	0.0464***	2.230***	2.386***	0.136***
	(0.0596)	(0.0497)	(0.0156)	(0.0348)	(0.0347)	(0.00775)
Obs	285,000	430,000	285,000	64,000	430,000	64,000
R-squared	0.1393	0.1964	0.09958	0.2055	0.2852	0.1188
Within R-squared	0.05035	0.09439	0.01323	0.1359	0.1549	0.02865

Notes: Regressions use tract-year-level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. Regression results are reported separately for the SBI (cols. 1-3) and high-tech (cols. 4-6) samples. The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table E.3: Neighboring tract controls: regression decomposition

	All Industries			Small Business Intensive			High-Tech		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	DHS(start. pc)	DHS(apps. pc)	Transition rate	DHS(start. pc)	DHS(apps. pc)	Transition rate	DHS(start. pc)	DHS(apps. pc)	Transition rate
Groups									
Demographic	0.023	0.004	0.023	0.021	0.001	0.015	0.011	0.024	0.005
Residual economic conditions	0.028	0.038	0.004	0.015	0.027	0.003	0.003	0.018	0.000
Incumbent firm characteristics	-0.010	-0.018	0.002	-0.002	0.000	0.002	0.003	0.006	0.003
Commercial share	0.171	0.233	0.007	0.091	0.113	0.003	0.026	0.024	0.001
Categories									
Local conditions	0.212	0.256	0.036	0.126	0.142	0.024	0.042	0.073	0.009
Neighborng tract conditions	0.020	0.034	0.010	0.011	0.023	0.006	0.009	0.028	0.005
Common market conditions	0.086	0.151	0.103	0.070	0.130	0.090	0.089	0.102	0.086
Residual	0.702	0.593	0.860	0.805	0.728	0.886	0.869	0.825	0.905

Notes: Decomposition of R^2 across own-tract variable groups for DHS(startups pc), DHS(apps pc), and transition rates for the all industries (cols. 1-3), small business intensive (cols. 4-6), and high-tech (cols. 7-9) regressions. Row 6 reports the contribution of all neighboring tract controls. "DHS" refers to the transformation based on Davis et al. (1996). "pc" refers to per capita (per 1,000 prime-age people).

E.2 Accounting for application characteristics in transition

At the transition rate margin, own-tract characteristics may partially related the composition and/or quality of ideas across locations. To evaluate this concern, we turn to an application-level analysis. At this level, we observe a set of characteristics, x, associated with each business application. These characteristics provide information about the type of business being applied for, and may also be informative about the quality of the underlying idea, ι.

Specifically, we examine the relationship between the probability of transition $p_l(\iota)$ in (2) and both local characteristics ($\mathbf{C}_l$) and application characteristics ($\mathbf{x}_{ilz}$) in a linear probability (LPM) framework:

$$p_{ilzt} = \mathbf{x}'_{ilzt}\boldsymbol{\gamma} + \mathbf{C}'_{lt-k}\boldsymbol{\beta^p} + f_{zt} + \epsilon^p_{ilzt}, \tag{15}$$

where $p_{ilzt} \in \{0, 1\}$ is an indicator of whether application i transitions.

The estimates of $\boldsymbol{\beta^p}$ are informative about the relationship between the transition of an idea and local characteristics, conditional on application characteristics (potential proxies for the quality of the idea). Going back to the model in Section 3, because locations differ in the distribution of ideas F_l, it is important to control for the composition of applications

in assessing the relationship between transition rate and local characteristics. The LPM in (Equation 15) does so by directly incorporating the individual application characteristics. We also compare the estimate of $\boldsymbol{\beta^p}$ with that of $\boldsymbol{\beta}$ based on Equation (14) to see whether controlling for application characteristics alters the general nature of the relationship between transition rate and local characteristics.

Our WBA sample includes over 2.3 million applications with substantially smaller number of applications for the high-tech and small business intensive industries. The control variables include the local condition variables used in the tract-level regressions, as well as a set of variables that control for application characteristics. The latter include controls for detailed industry of the application (4-digit NAICS), reason for applying (planned wages, obtaining a business bank account), type of entity applying (e.g., corporation, partnership), LLC status, inclusion of a trade name, business start date, and quarter of application.[44] The models include county-by-year fixed effects, and standard errors are clustered at the county level.

The results are presented in Table E.4, where only the coefficients of the local conditions are shown.[45] Higher values of local demographic and household economic variables, such as African American share, foreign born share and the unemployment rate are associated with a lower probability of transitioning to employer status across all three industry samples. Meanwhile, a higher share of the population with at least a bachelors degree and higher house values are associated with a higher likelihood of transitioning. Across all three industry samples, locations with a higher share of employment in large firms or greater average size have a lower likelihood of becoming an employer business, whereas the commercial share variable is positively associated with transitioning. These findings are broadly consistent with the tract-level results on transition rates reported in Tables 5, D.4 and D.7 and indicate that application success, as measured by the transition to employer-business status, is systematically related to local conditions, even after controlling for application heterogeneity.

[44] See Bayard et al. (2018) for a detailed discussion of estimating transition probabilities based on application characteristics. Our analysis here follows closely that approach with two caveats. The model estimated here includes detailed tract-level characteristics but excludes the detailed interactions among variables. Moreover, our model is estimated pooled across states, whereas in Bayard et al. (2018) the models are estimated individually at the state level with a rich set of industry-application characteristic interactions.

[45] The coefficients on the application characteristics are omitted as they involve a large number of estimates which are not our focus.

Table E.4: Tract-level linear probability model (LPM) estimates

	(1) All Industries	(2) SBI	(3) High-Tech
log(median age)	0.0170**	0.0108	–0.00815
	(0.00669)	(0.00847)	(0.0143)
bachelors or higher share	0.0401***	0.0488***	0.0920***
	(0.0091)	(0.0122)	(0.0307)
some college share	–0.0383***	–0.0270*	0.0322
	(0.0115)	(0.0153)	(0.0446)
African American share	–0.153***	–0.190***	–0.137***
	(0.0064)	(0.00678)	(0.0176)
Asian share	0.019	–0.0171	0.0484*
	(0.0187)	(0.0186)	(0.0271)
Hispanic share	–0.0191	–0.0281**	–0.0393
	(0.0153)	(0.0139)	(0.0249)
foreign born share	–0.0523***	–0.0495***	–0.0667**
	(0.0145)	(0.0142)	(0.0297)
log(per capita income)	–0.00819**	–0.00594	–0.0207**
	(0.00342)	(0.00438)	(0.00967)
unemployment rate	–0.0458***	–0.0451**	–0.0527
	(0.0143)	(0.0176)	(0.0477)
log(median house value)	0.0254***	0.0304***	0.0231***
	(0.00196)	(0.00258)	(0.00621)
share of emp in young firms	0.0131**	0.0105	–0.0153
	(0.00515)	(0.00796)	(0.02)
share of emp in large firms	–0.00835**	–0.00367	0.00917
	(0.00373)	(0.00577)	(0.0142)
DHS(avg firm emp)	–0.00773***	–0.00862***	–0.00936**
	(0.00105)	(0.00166)	(0.00455)
commercial share	0.0733***	0.0676***	0.0288**
	(0.00324)	(0.00475)	(0.0142)
Obs	2,355,000	97,080	737,200
ind_share	NAICS3	NAICS3	NAICS3
fe	fips x yr	fips x yr	fips x yr
cluster	fips	fips	fips
R-squared	0.1136	0.1099	0.1325
Within R-squared	0.00793	0.0105	0.006626

Notes: Regressions use application-level data, control for 3-digit NAICS industry shares of tract employment, and include county-by-year fixed effects. The observation counts have been rounded for disclosure reasons. Standard errors are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

F Estimated elasticities and magnitudes

F.1 Elasticities

Consider the relationship in Equation (14). The transformation of startups per capita is

$$\widetilde{S}_{lzt} = 2\left(\frac{S_{lzt} - \overline{S}}{S_{lzt} + \overline{S}}\right), \tag{16}$$

where $\overline{S}$ is the grand mean of S_{lzt} over all locations l (and z where l belongs to) and time periods t:

$$\overline{S} = \frac{1}{N} \sum_t \sum_l S_{lzt}. \tag{17}$$

Define $\widetilde{A}_{lzt}$ analogously for applications per capita, A_{lzt}. The transition rate T_{lzt} is left untransformed.

To obtain the point elasticity of S_{lzt} with respect to a covariate c_{lt-k} (expressed in levels), differentiate (16). Note that $\overline{S}$ depends on S_{lzt} and therefore on c_{lt-k}. Under the linear model, however, S_{lzt} does not depend on $c_{l't'-k}$ for $l' \neq l$ or $t' \neq t$, implying

$$\frac{\partial \overline{S}}{\partial c_{lt-k}} = \frac{1}{N} \frac{\partial S_{lzt}}{\partial c_{lt-k}}.$$

Differentiating (16) yields

$$\begin{aligned}
\frac{\partial}{\partial c_{lt-k}} 2 \left(\frac{S_{lzt} - \overline{S}}{S_{lzt} + \overline{S}} \right) &= 2 \frac{\frac{\partial S_{lzt}}{\partial c_{lt-k}} \left[\left(1 - \frac{1}{N}\right) (S_{lzt} + \overline{S}) - \left(1 + \frac{1}{N}\right) (S_{lzt} - \overline{S}) \right]}{(S_{lzt} + \overline{S})^2} \\
&= -\frac{4}{N} \frac{S_{lzt} - N\overline{S}}{(S_{lzt} + \overline{S})^2} \frac{\partial S_{lzt}}{\partial c_{lt-k}} \\
&= \beta_c^S,
\end{aligned}$$

where the last equality follows from the regression model. Thus, the elasticity of the original variable is

$$\begin{aligned}
\epsilon_c^S(S_{lzt}, c_{lt-k}) &= \frac{\partial S_{lzt}}{\partial c_{lt-k}} \frac{c_{lt-k}}{S_{lzt}} \\
&= \frac{(S_{lzt} + \overline{S})^2}{-\frac{4}{N}(S_{lzt} - N\overline{S})} \beta_c^S \frac{c_{lt-k}}{S_{lzt}}.
\end{aligned}$$

For large N (as in our application, where the (l,t) sample is very large),

$$-\frac{4}{N}(S_{lzt} - N\overline{S}) \simeq 4\overline{S},$$

so that an approximation for the elasticity is

$$\epsilon_c^S(S_{lzt}, c_{lt-k}) \simeq \frac{1}{4}\beta_c^S \frac{(S_{lzt}+\overline{S})^2}{\overline{S}} \frac{c_{lt-k}}{S_{lzt}}.$$

Similarly, for applications per capita,

$$\epsilon_c^A(A_{lzt}, c_{lt-k}) \simeq \frac{1}{4}\beta_c^A \frac{(A_{lzt}+\overline{A})^2}{\overline{A}} \frac{c_{lt-k}}{A_{lzt}}.$$

For the untransformed transition rate,

$$\epsilon_c^T(T_{lzt}, c_{lt-k}) = \beta_c^T \frac{c_{lt-k}}{T_{lzt}}.$$

Evaluating each elasticity at the sample means gives

$$\begin{aligned}
\epsilon_c^S(\overline{S}, \overline{c}) &\simeq \frac{1}{4}\beta_c^S \frac{(2\overline{S})^2}{\overline{S}} \frac{\overline{c}}{\overline{S}} = \beta_c^S\, \overline{c}, \\
\epsilon_c^A(\overline{A}, \overline{c}) &\simeq \beta_c^A\, \overline{c}, \\
\epsilon_c^T(\overline{T}, \overline{c}) &= \beta_c^T \frac{\overline{c}}{\overline{T}}.
\end{aligned}$$

Some covariates enter the model in logs or are themselves transformed via (16). For a log covariate,

$$\begin{aligned}
\epsilon_c^S(\overline{S}, \overline{c}) &\simeq \beta_c^S, \\
\epsilon_c^A(\overline{A}, \overline{c}) &\simeq \beta_c^A, \\
\epsilon_c^T(\overline{T}, \overline{c}) &= \beta_c^T \frac{1}{\overline{T}}.
\end{aligned}$$

For a covariate transformed using (16),

$$2\frac{\partial}{\partial c_{lt-k}}\left(\frac{S_{lzt}-\overline{S}}{S_{lzt}+\overline{S}}\right) = 2\beta_c^S \frac{\partial}{\partial c_{lt-k}}\left(\frac{c_{lt-k}-\overline{c}}{c_{lt-k}+\overline{c}}\right),$$

which implies

$$\frac{S_{lzt}-N\overline{S}}{(S_{lzt}+\overline{S})^2}\frac{\partial S_{lzt}}{\partial c_{lt-k}}\frac{c_{lt-k}}{S_{lzt}} = \beta_c^S \frac{c_{lt-k}-N\overline{c}}{(c_{lt-k}+\overline{c})^2}\frac{c_{lt-k}}{S_{lzt}}.$$

Hence,

$$\epsilon_c^S(S_{lzt}, c_{lt-k}) = \beta_c^S \frac{(S_{lzt} + \overline{S})^2}{S_{lzt} - N\overline{S}} \frac{c_{lt-k} - N\overline{c}}{(c_{lt-k} + \overline{c})^2} \frac{c_{lt-k}}{S_{lzt}},$$

and at the means,

$$\epsilon_c^S(\overline{S}, \overline{c}) \simeq \beta_c^S.$$

A similar argument yields

$$\epsilon_c^A(\overline{A}, \overline{c}) \simeq \beta_c^A.$$

Finally, for the transition rate with a DHS-transformed covariate,

$$\begin{aligned}\epsilon_c^T(T_{lzt}, c_{lt-k}) &= \frac{\partial T_{lzt}}{\partial c_{lt-k}} \frac{c_{lt-k}}{T_{lzt}} \\ &= 2\beta_c^T \frac{\partial}{\partial c_{lt-k}} \left(\frac{c_{lt-k} - \overline{c}}{c_{lt-k} + \overline{c}} \right) \frac{c_{lt-k}}{T_{lzt}},\end{aligned}$$

and at the means,

$$\epsilon_c^T(\overline{T}, \overline{c}) \simeq \beta_c^T \frac{1}{\overline{T}},$$

where the last approximation uses the same large-N simplification as above.

Thus, the elasticities can be estimated by replacing the population parameters $(\beta_c^S, \beta_c^A, \beta_c^T)$ with their sample estimates, yielding $\widehat{\epsilon}_c^Y(\overline{Y}, \overline{c})$ for $Y \in \{S, A, T\}$.

F.2 Magnitudes

For any covariate c, the estimated percent change in $Y \in \{S, A, T\}$ induced by a one–standard-deviation change in c (expressed in percent of its mean) is

$$\widehat{\Delta Y_l} = \widehat{\epsilon}_c^Y(\overline{Y}, \overline{c}) \left(100 \times \frac{s_c}{m_c} \right) = \widehat{\epsilon}_c^Y(\overline{Y}, \overline{c}) \left(100 \times CV_c \right),$$

where s_c, m_c, and $CV_c = s_c/m_c$ denote the sample standard deviation, mean, and coefficient of variation of the untransformed covariate c, respectively.